

Gut Microbe-Derived Trimethylamine Shapes Circadian Rhythms Through the Host Receptor TAAR5

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eLife Assessment

This study presents an **important** finding linking the bacterial metabolite trimethylamine and its receptor to circadian rhythms and olfaction. The current evidence supporting the claims of the authors is **convincing**, although further data and improvements to the presentation would further increase the impact of these results. This work will be of broad interest to researchers interested in nutrition, microbial metabolism, circadian rhythms, and host-microbiome interactions.

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Abstract

Elevated levels of the gut microbe-derived metabolite trimethylamine N-oxide (TMAO) are associated with cardiometabolic disease risk. However, the mechanism(s) linking TMAO production to human disease are incompletely understood. Initiation of the metaorganismal TMAO pathway begins when dietary choline and related metabolites are converted to trimethylamine (TMA) by gut bacteria. Gut microbe-derived TMA can then be further oxidized by host flavin-containing monooxygenases to generate TMAO. Previously, we showed that drugs lowering both TMA and TMAO protect mice against obesity via rewiring of host circadian rhythms. Although most mechanistic studies in the literature have focused on the metabolic end product TMAO, here we have instead tested whether the primary metabolite TMA alters host metabolic homeostasis and circadian rhythms via trace amine-associated receptor 5 (TAAR5). Remarkably, mice lacking the host TMA receptor (*Taar5*^{-/-})

have altered circadian rhythms in gene expression, metabolic hormones, gut microbiome composition, and innate behaviors. In parallel, mice genetically lacking bacterial TMA production or host TMA oxidation have altered circadian rhythms. These results provide new insights into diet-microbe-host interactions relevant to cardiometabolic disease.

Highlights

- Mice lacking the host TMA receptor (*Taar5*^{-/-}) have altered circadian rhythms.
- *Taar5*^{-/-} mice have altered innate behaviors in a time of day dependent manner.
- The normal circadian oscillations in the gut microbiome are dysregulated in *Taar5*^{-/-} mice.
- Genetic deletion of bacterial TMA production or host TMA oxidation shapes circadian rhythms.

Impact statement

The metaorganismal TMAO pathway is strongly associated with cardiometabolic disease, yet mechanisms underlying this are incompletely understood. Mahen *et al.* now show that microbial TMA production and host TMA sensing via TAAR5 broadly alter host circadian rhythms in gene expression, metabolic hormones, gut microbiome, and innate behaviors.

Introduction

Modern investigation into human disease relies heavily on genetic and genomic approaches, and most assume that human disease is primarily driven by variation in the human genome. However, in the post genomic era we now understand that human genetic variation explains only a small proportion of risk for complex diseases such as obesity, diabetes, and cardiovascular disease (CVD). Instead, in many cases environmental factors are the predominant drivers of cardiometabolic disease pathogenesis. Among all contributing environmental factors, it is clear that dietary patterns, and diet-driven alterations in the gut microbiome, can profoundly impact many common human diseases.^{1,2,3,4} There are now many examples of diet-microbe-host interactions shaping human disease, but one of the more compelling is the reproducible link between elevated trimethylamine N-oxide (TMAO) levels and cardiovascular disease (CVD) risk.^{5,6,7,8,9,10,11,12} TMAO is generated by a metaorganismal (*i.e.* microbe & host) pathway where dietary substrates such as choline, *L*-carnitine, and ψ -butyrobetaine are metabolized by gut microbial enzymes to generate the primary metabolite trimethylamine (TMA).^{5,6,7,8,9,10,11,12} TMA is then further metabolized by the host enzyme flavin-containing monooxygenase 3 (FMO3) in the liver to produce trimethylamine-N-oxide (TMAO).^{10,11} Elevated TMAO levels are associated with many human diseases including diverse forms of CVD^{5,6,7,8,9,10,11,12}, obesity,^{13,14,15} type 2 diabetes,^{15,16,17} chronic kidney disease (CKD),^{17,18,19} neurodegenerative conditions including Parkinson's and Alzheimer's disease,^{19,20,21} and several cancers.^{21,22,23} Many of these disease associations have been validated in several large population meta-analyses^{23,24,25} and Mendelian randomization studies.^{26,27,28} The emerging body of evidence supports the notion that elevated TMAO levels are causally related to cardiometabolic disease pathogenesis. In further support, therapies aimed at lowering circulating TMAO levels provide striking protection against cardiometabolic disease in animal models.^{28,29,30,31,32,33,34,35,36} Several independent studies have now shown that inhibition of either the host TMAO-producing enzyme FMO3 or the microbial TMA-producing

enzyme CutC protects against diet-induced atherosclerosis^{15,30}, heart failure³², thrombosis^{29,31}, obesity^{13,33}, liver disease³⁴, insulin resistance^{13,33}, CKD^{35,36}, and abdominal aortic aneurysm³⁷.

Even though TMAO lowering therapies are very effective in preclinical animal models, the underlying mechanism(s) linking the metaorganismal production of TMAO to cardiometabolic disease pathogenesis are still incompletely understood. There is some evidence that TMAO can promote inflammatory processes via activation of the nucleotide-binding domain, leucine-rich-containing family, pyrin domain-containing-3 (NLRP3) inflammasome and nuclear factor κ B (NF κ B)^{38–41}. TMAO can also activate the endoplasmic reticulum (ER) stress kinase PERK (EIF2AK3) in hepatocytes to promote metabolic disturbance²⁸. In parallel, TMAO promotes stimulus-dependent calcium release in platelets to promote thrombosis⁸. Although the end product of the pathway TMAO clearly impacts cell signaling in the host to impact cardiometabolic disease, these TMAO-driven mechanisms do not fully explain how elevated TMAO levels contribute to so many diverse diseases in humans. In addition to TMAO-driven signaling mechanisms, we recently reported that major components of the TMAO pathway (choline, TMA, FMO3, and TAAR5) oscillate in a highly circadian fashion³³. Furthermore, gut microbe-targeted drugs that selectively block TMA production alter host circadian rhythms in the gut microbiome and host phospholipid metabolism³³. It is important to note that disruption of the circadian clock is a common hallmark of almost all diseases where TMAO levels are elevated^{42–45}. To follow up on the potential links between the TMAO pathway and host circadian disruption here we have used genetic knockout approaches at the level of gut microbial TMA production (i.e. *cutC*-null microbial communities), host TMA oxidation (i.e. *Fmo3*^{-/-}), or host sensing of TMA (i.e. *Taar5*^{-/-}). Results here further bolster the concept that TMA-driven TAAR5 activation shapes host circadian rhythms.

Results

Mice Lacking the TMA Receptor TAAR5 Have Altered Circadian Rhythms in Metabolic Homeostasis and Innate Olfactory-Related Behaviors

We recently demonstrated that drugs blocking gut microbial TMA production protect against obesity via rewiring circadian rhythms in the gut microbiome, liver, white adipose tissue and skeletal muscle³³. Furthermore, we also showed that blocking bacterial TMA production elicited unexpected alterations in olfactory perception of diverse odorant stimuli⁴⁶. Therefore, we have followed up here to further interrogate circadian rhythms in the gut microbiome, liver, white adipose tissue (WAT), skeletal muscle, and olfactory bulb in mice lacking the only known host G protein-coupled receptor (GPCR) that senses TMA known as TAAR5^{47,48}. In agreement with our previous report showing that *Taar5* mRNA is expressed in a circadian manner in skeletal muscle³³, *LacZ* reporter expression oscillates with highest expression in the dark cycle in muscle (**Figure 1A**). However, unlike the striking impact that choline TMA lyase inhibitors have on the core circadian clock machinery in skeletal muscle³³, mice lacking the TMA receptor (*Taar5*^{-/-}) have unaltered circadian gene expression in muscle (**Figure 1A**). Instead, *Taar5*^{-/-} mice have alterations in the expression of key circadian genes including *Arntl*, *Clock*, *Nr1d1*, *Cry1*, and *Per2* in the olfactory bulb (**Figure 1B**). There is also some reorganization of circadian gene expression in the liver (**Figure 1C**) and gonadal white adipose tissue (**Figure 1D**), albeit modest. Given the key role that the TMAO pathway plays in suppressing the beiging of white adipose tissue¹³, it is important to note that circadian oscillation in genes related to WAT beiging including *Prdm16* and *Ucp1* was altered in *Taar5*^{-/-} mice (**Figure 1D**). We next examined circadian oscillations in circulating metabolite, hormone, and cytokine levels in *Taar5*^{+/+} and *Taar5*^{-/-} mice (**Figure S1**). When we measured substrates for gut microbial TMA production we

found that *Taar5*^{-/-} mice had normal oscillations in choline and ψ -butyrobetaine, but slightly altered plasma levels of *L*-carnitine particularly in the dark cycle (**Figure S1A**). *Taar5*^{-/-} mice also had increased levels of TMA at ZT22, when compared to *Taar5*^{+/+} controls (**Figure S1A**). Interestingly, *Taar5*^{-/-} mice had altered rhythmic levels in host metabolic hormones including insulin, peptide YY (PYY), leptin, ghrelin, and active glucagon-like peptide 1 (GLP-1) (**Figure S1B**) and modest differences in circulating cytokines including monocyte chemoattractant 1 (MCP-1), interleukin 6 (IL-6), and tumor necrosis factor α (TNF α) (**Figure S1B**). Collectively, these data demonstrate that *Taar5*^{-/-} mice have altered circadian-related gene expression that is most apparent in the olfactory bulb (**Figure 1**), and abnormal circadian oscillations in circulating hormones and cytokines (**Figure S1**).

We next set out to comprehensively analyze the circadian rhythms in behavioral phenotypes in mice lacking the TMA receptor TAAR5. In this line of investigation we took a very broad approach to examine impacts of *Taar5* deficiency on metabolic, cognitive, motor, anxiolytic, social, olfactory and innate behaviors. The main rationale behind this in-depth investigation was to allow for comparison to our recent work showing that pharmacologic blockade of the production of the TAAR5 ligand TMA (using choline TMA lyase inhibitors) produced clear metabolic, innate, and olfactory-related social behavioral phenotypes, but did not dramatically impact aspects of cognition, motor, or anxiolytic behaviors.^{33,46} Also, it is important to note other groups have recently shown that TAAR5 activation with non-TMA ligands or genetic deletion of *Taar5* results in clear olfactory^{48–51}, anxiolytic⁵¹, cognitive⁵², and sensorimotor^{53,54} behavioral abnormalities. Here, it was our main goal to identify whether any behavioral phenotypes that were consistently seen in mice lacking bacterial TMA production^{33,46} or host TMA sensing by TAAR5 (studied here) are time-of-day dependent indicating circadian inputs. Upon comprehensive behavioral phenotyping, it became very clear that many behavioral phenotypes in *Taar5*^{-/-} are in most cases gender-specific. For example, only in females we found that startle test response was enhanced, forepaw grip strength was reduced, freezing response during fear conditioning was reduced, and oxygen consumption during cold-induced thermogenesis was increased in *Taar5*^{-/-} mice compared to controls (**Figures S2A**, **S2B**, **S2F**, and **S2J**). There were also examples of behavioral differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice that were most apparent with both sexes combined (**Figures S2C**, **S2D**, and **S2E**). Also, it is important to note that many behavioral phenotypes examined, including tests not shown, were unaltered in *Taar5*^{-/-} mice (**Figures S2G**, **S2H**, and **S2I**). Given the consistent alterations in innate and olfactory-related phenotypes seen in both mice lacking bacterial TMA production^{33,46} and mice lacking host TMA sensing by TAAR5 here, we next followed up to perform select innate and olfactory-related behavioral tests in mice at defined circadian time points (**Figure 2**). To study the circadian presentation of phenotype, we carefully controlled the time window of testing for either the olfactory cookie test or marble burying test, both of which have been shown to be altered in mice lacking bacterial TMA synthesis.^{46,55} When the olfactory cookie test was performed during the mid light cycle (ZT5–ZT7), the latency to find the buried cookie was significantly increased in male *Taar5*^{-/-} mice compared to *Taar5*^{+/+} controls (**Figure 2A**). However, when the same mice performed the olfactory cookie test at the dark-to-light phase transition (ZT23–ZT1), or at the light-to-dark phase transition (ZT13–ZT15), there was no significant differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice (**Figure 2A**). When subjected to the marble burying test, only female *Taar5*^{-/-} mice buried significantly more marbles than female wild type controls only at ZT5–ZT7, but this was not apparent at other ZT timepoints (**Figure 2B**). Collectively, these data demonstrate that *Taar5*^{-/-} mice exhibit highly gender-specific alterations in innate and olfactory behaviors, and these behavior phenotypes are only apparent at certain periods within the light cycle (**Figures 2**, and **S2**).

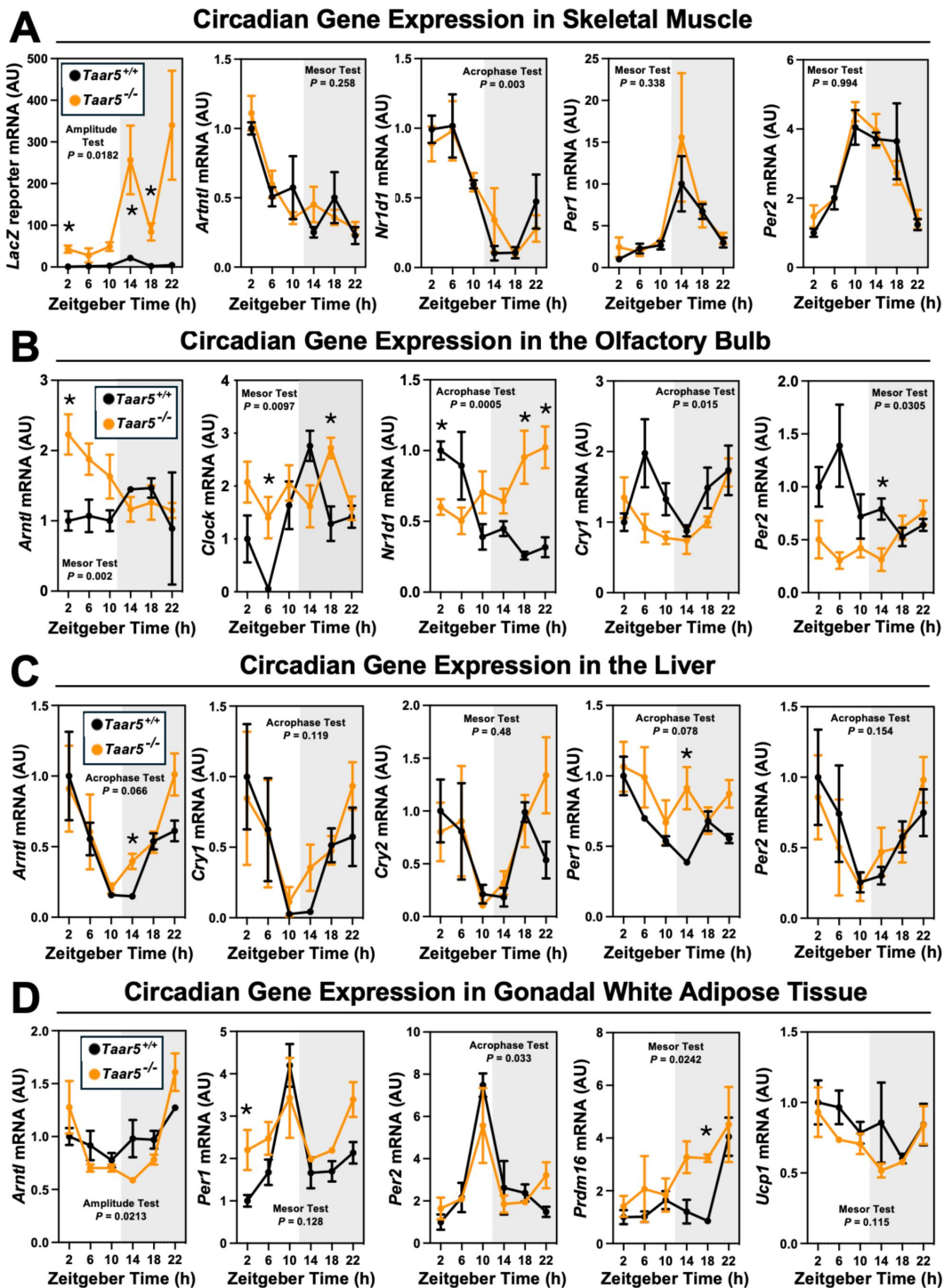


Figure 1.

The Host Trimethylamine Receptor TAAR5 Shapes Tissue-Specific Circadian Oscillations.

Male chow-fed wild type (*Taar5*^{+/+}) or mice lacking the TMA receptor (*Taar5*^{-/-}) were necropsied at 4-hour intervals to collect tissues including skeletal muscle (**A**), olfactory bulb (**B**), liver (**C**), or gonadal white adipose tissue (**D**). The relative gene expression for circadian and metabolism related genes was quantified by qPCR. Data shown represent the means $\pm$ S.D. for $n=3-6$ individual mice per group. Group differences were determined using cosinor analyses p -values are provided where there were statistically significant differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice. The complete cosinor statistical analysis for circadian data can be found in Table S1. * = Significant differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice by student's t -tests within each ZT time point ($p<0.05$).

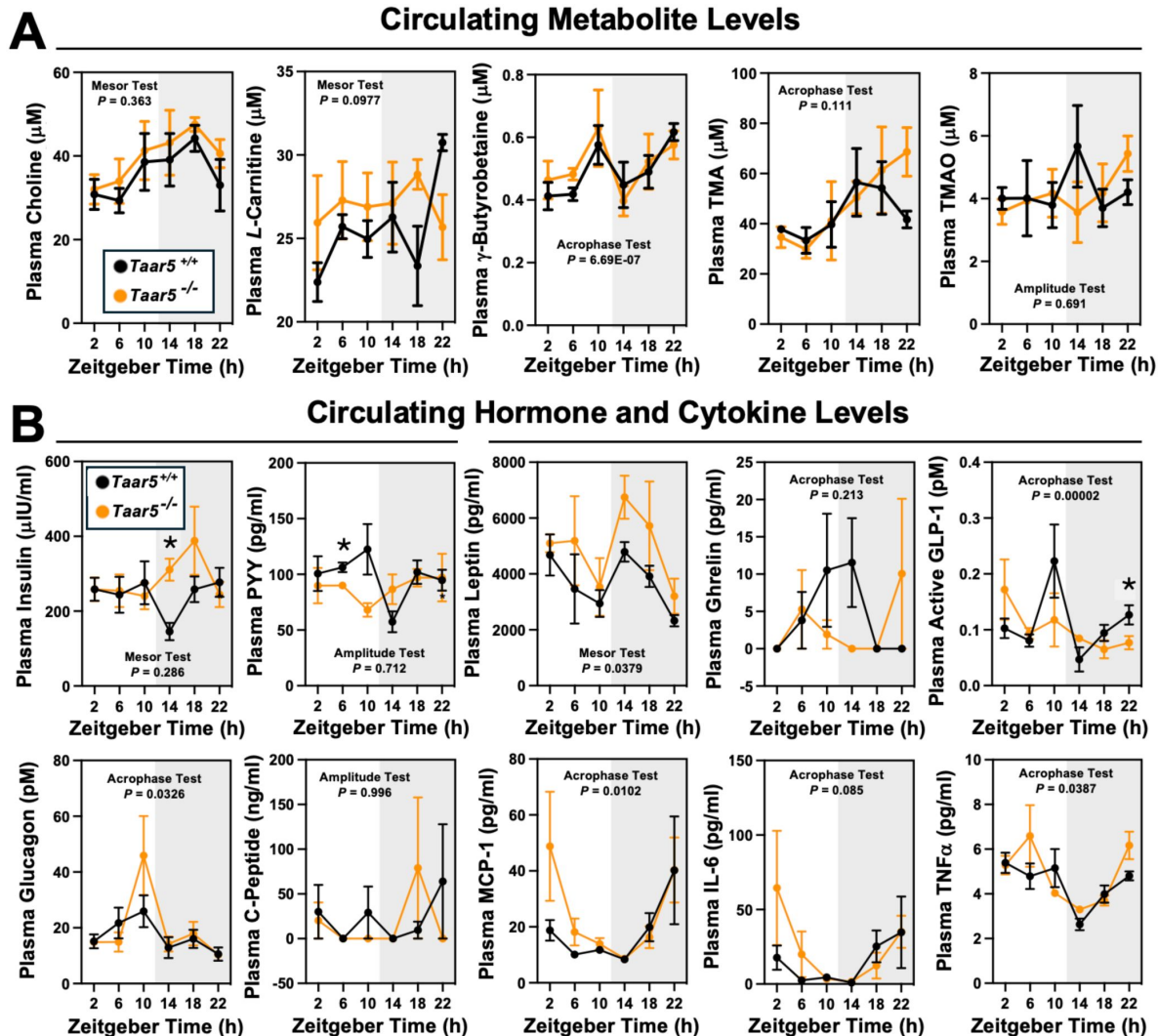


Figure 1 – Figure Supplement 1.

The Host Trimethylamine Receptor TAAR5 Shapes Circadian Oscillations in Circulating Hormones and Cytokines.

Male chow-fed wild type mice (*Taar5*^{+/+}) or mice lacking the TMA receptor (*Taar5*^{-/-}) were necropsied at 4-hour intervals to collect plasma. **(A)** Plasma levels of metabolites in the metaorganismal trimethylamine N-oxide (TMAO) pathway were quantified using liquid chromatography tandem mass spectrometry (LC-MS/MS). **(B)** Plasma hormones and cytokine levels were quantified as described in the material and methods. Data shown represent the means $\pm$ S.D. for $n = 3$ -6 individual mice per group. Group differences were determined using cosinor analyses p -values are provided where there were statistically significant differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice. The complete cosinor statistical analysis for circadian data can be found in Table S1. * = Significant differences between *Taar5*^{+/+} and *Taar5*^{-/-} mice by student's t -tests within each ZT time point ($p < 0.05$).

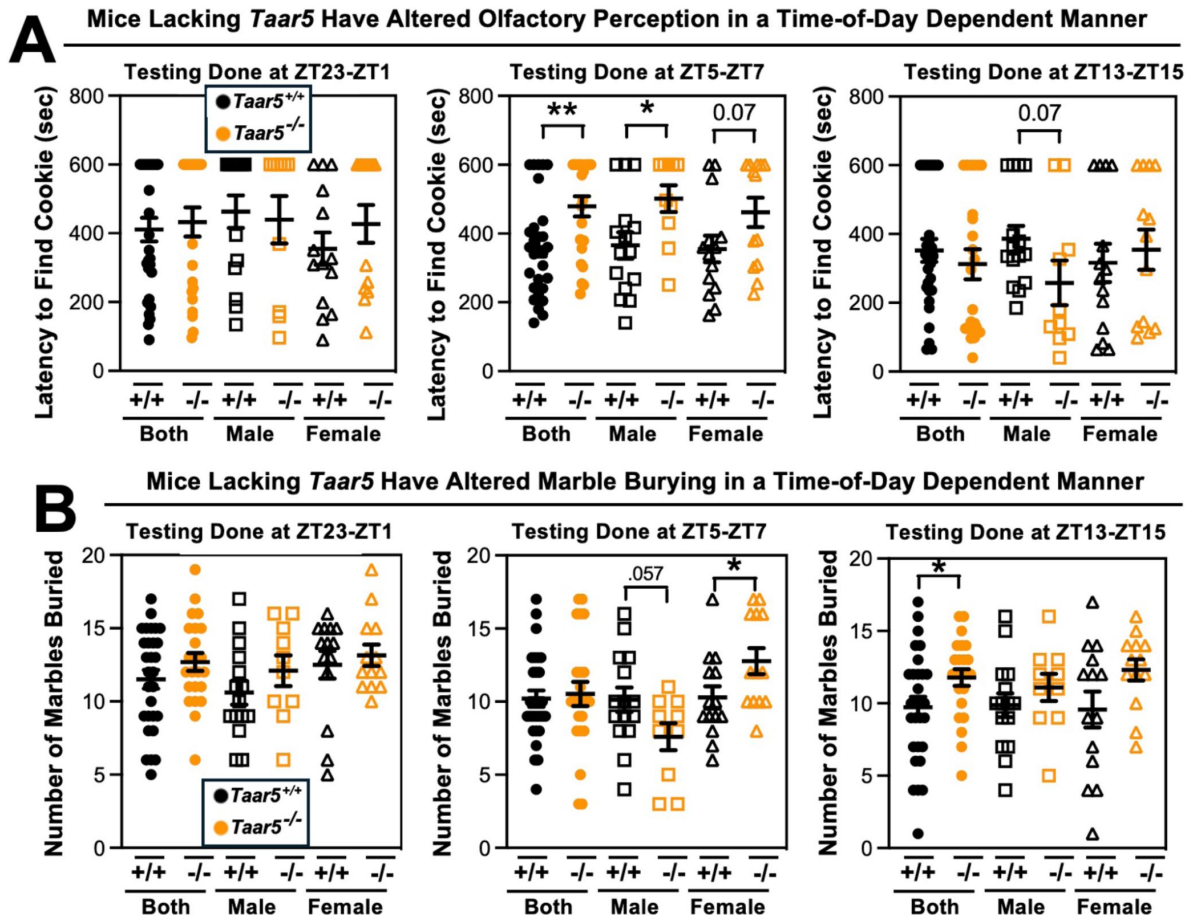


Figure 2.

Mice Lacking the Host TMA Receptor TAAR5 Have Altered Olfactory and Repetitive Behaviors Only at Specific Circadian Timepoints.

Male or female wild type mice (*Taar5*^{+/+}) or mice lacking the TMA receptor (*Taar5*^{-/-}) were subjected to the olfactory cookie test (**A**) or the marble burying test (**B**). To examine circadian alterations in behavior, these tests were done in either the dark-light phase transition (ZT23-ZT1), mid light cycle (ZT5-ZT7), or early dark cycle (ZT13-ZT15). Data represent the mean $\pm$ S.E.M., and statistically significant difference between *Taar5*^{+/+} and *Taar5*^{-/-} mice are denoted by * = $p < 0.05$ and ** = $p < 0.01$.

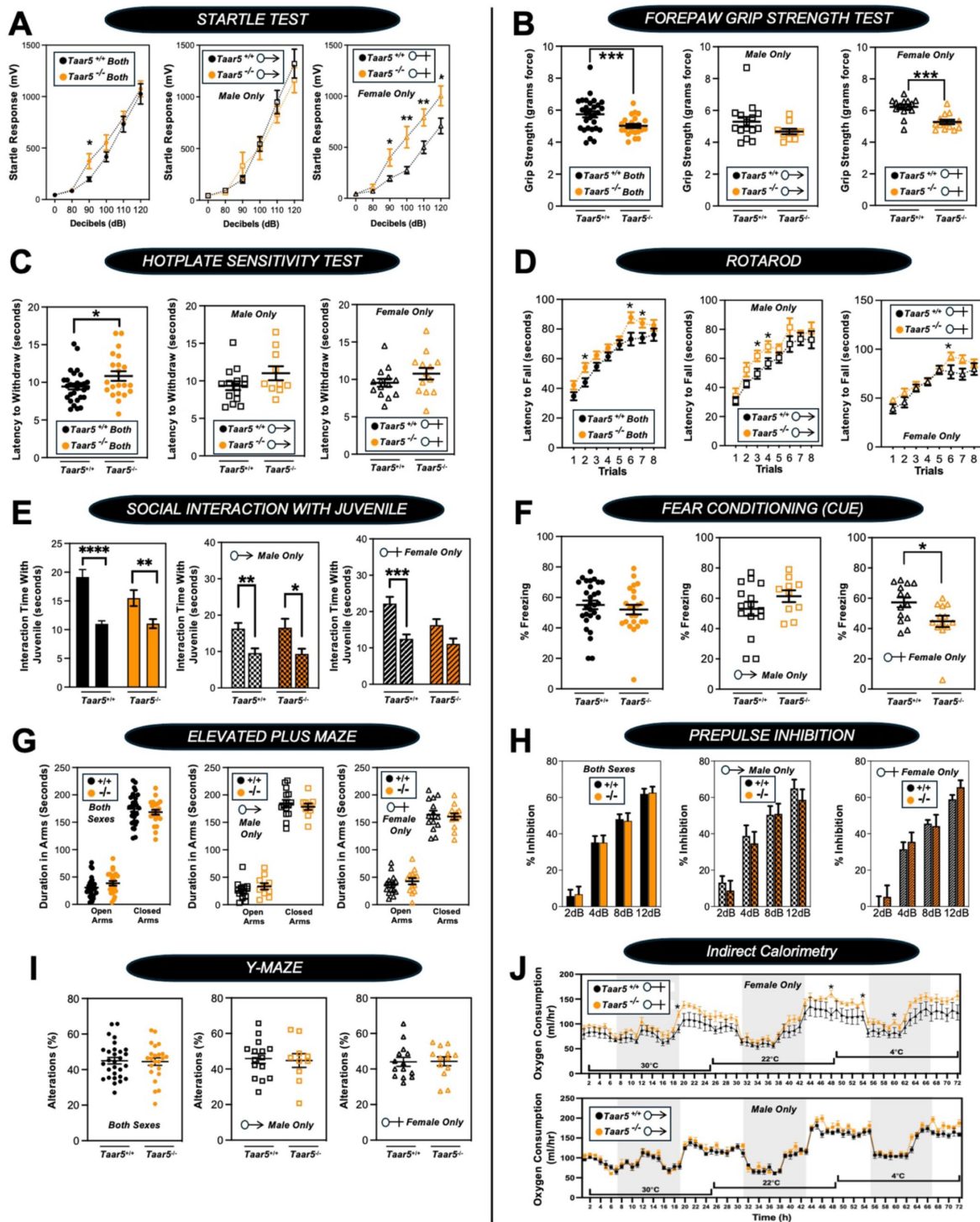


Figure 2 - Figure Supplement 1.

Mice Lacking the Host TMA Receptor TAAR5 Have Altered Behaviors.

Male or female wild type mice ($Taar5^{+/+}$) or mice lacking the TMA receptor ($Taar5^{-/-}$) were subjected to a battery of behavioral test including: (A) startle test, (B) forepaw grip strength test, (C) hotplate sensitivity test, (D) rotarod test, (E) social interaction with a juvenile test, with initial testing and then again three days later. (F) fear conditioning cue test, (G) elevated plus maze, (H) prepulse inhibition, (I) Y-maze, and (J) indirect calorimetry using the Oxymax CLAM metabolic cage system. Data are shown for the entire cohort combining both sexes or divided into either male or female cohorts to examine sexual dimorphism in phenotype. Data represent the mean $\pm$ S.E.M., and statistically significant difference between $Taar5^{+/+}$ and $Taar5^{-/-}$ mice are denoted by * = $p < 0.05$ and ** = $p < 0.01$.

Host TAAR5 Regulates the Circadian Rhythmicity of the Gut Microbiome

Bi-directional microbe-host communication is required for homeostatic control of chronobiology in the metaorganism.^{56–60} Our data strongly suggest that gut microbe-derived TMA can shape host circadian rhythms in metabolic homeostasis and behavior^{33,46}, but we also wanted to test whether host TAAR5 may reciprocally regulate circadian rhythms of the gut microbiome^{58,59}. When we examined the cecal microbiome in *Taar5*^{+/+} and *Taar5*^{-/-} mice, there were striking alterations that were time of day dependent (**Figures 3** and **S3**). In particular, the oscillatory behavior of several *Lachnospiraceae*, *Odoribacter*, and *Dubosiella* genera are altered in *Taar5*^{-/-} mice when compared to wild type controls (**Figures 3** and **S3**). Although there were many microbiome alterations, *Taar5*^{-/-} mice showed loss of the normal rhythmicity for *Dubosiella* and *Odoribacter* genera yet gained in amplitude of rhythmicity for *Bacteroides* genera (**Figure 3** and **S3**). It is important to note that previous independent studies have also shown that either blocking bacterial TMA synthesis³³ or FMO3-driven TMA oxidation²⁹ can also strongly reorganize the gut microbiome in mice. Although more work is needed to fully understand the underlying mechanisms, it is clear that both gut microbe-driven TMA production and host sensing of TMA by TAAR5 (**Figures 3** and **S3**) can strongly impact on circadian oscillations of the gut microbiome.

Mice Genetically Lacking Either Gut Microbial TMA Production or Host-Driven TMA Oxidation Have Altered Circadian Rhythms

To confirm and extend the idea that gut microbe-derived TMA can shape host circadian rhythms we next performed experiments where we genetically deleted either gut microbial TMA synthesis or host-driven TMA oxidation. First, we used gnotobiotic mice engrafted with a defined microbial community with or without genetic deletion of the choline TMA lyase CutC (i.e. *Clostridium sporogenes* wild type versus $\Delta cutC$) to understand the ability of gut microbe-derived TMA to alter host circadian rhythms (**Figure 4** and **S4**). Circadian oscillations in plasma TMA and TMAO are only detectable in the group colonized with *C. sporogenes* (WT), but not $\Delta cutC$ *C. sporogenes* (**Figure 4A**). Other TMAO pathway-related metabolites including choline, L-carnitine, ψ -butyrobetaine, and betaine exhibit very modest alterations in circadian rhythms (**Figure 4A**). In a similar manner to *Taar5*^{-/-} mice (**Figure 1**), mice colonized with $\Delta cutC$ *C. sporogenes* show clear alterations in the expression of key circadian genes including *Arntl*, *Clock*, *Nr1d1*, *Cry1*, *Per1*, and *Per2* in the olfactory bulb (**Figure 4B**). In line with our previous findings that FMO3 suppresses both beige and brown fat-induced cold induced thermogenesis¹³, we also found altered expression of key circadian genes (*Arntl*, *Cry1*, *Per2*) and metabolic genes (*Pemt* and *Pdk4*) in subscapular brown adipose tissue (BAT) of $\Delta cutC$ *C. sporogenes*-colonized mice (**Figure S4A**). Furthermore, mice colonized with $\Delta cutC$ *C. sporogenes* have altered circadian oscillations in circulating cytokines/chemokines (IL-1 α , IL-2, IL-33, IFN ψ , and MCP-1), metabolic hormones (leptin and glucagon-like peptide 1), amino acids (tryptophan), and diverse microbe-associated metabolites (indole acetic acid, phenylacetic acid, and phenylacetylglutamine) (**Figures 4C** and **S4B**). Although gut microbes are the sole source of TMA, circulating levels are also shaped by abundant conversion of TMA to TMAO by the host liver enzyme FMO3^{13,29}. To test whether the hepatic conversion of TMA to TMAO by FMO3 may alter circadian rhythms, we necropsied female *Fmo3*^{+/+} and *Fmo3*^{-/-} mice at ZT2 (early light cycle) or ZT14 (early dark cycle). *Fmo3*^{-/-} mice have reduced levels of TMAO at both ZT2 and ZT14, as well as reduced expression of key circadian genes *Arntl* and *Per1* at ZT2 (**Figure S4C**). Taken together, our data suggest that bacterial production, FMO3-driven metabolism of TMA, as well as sensing of TMA by the host receptor TAAR5, converge to shape circadian rhythms in gene expression, metabolic hormones, gut microbiome composition, and innate behaviors.

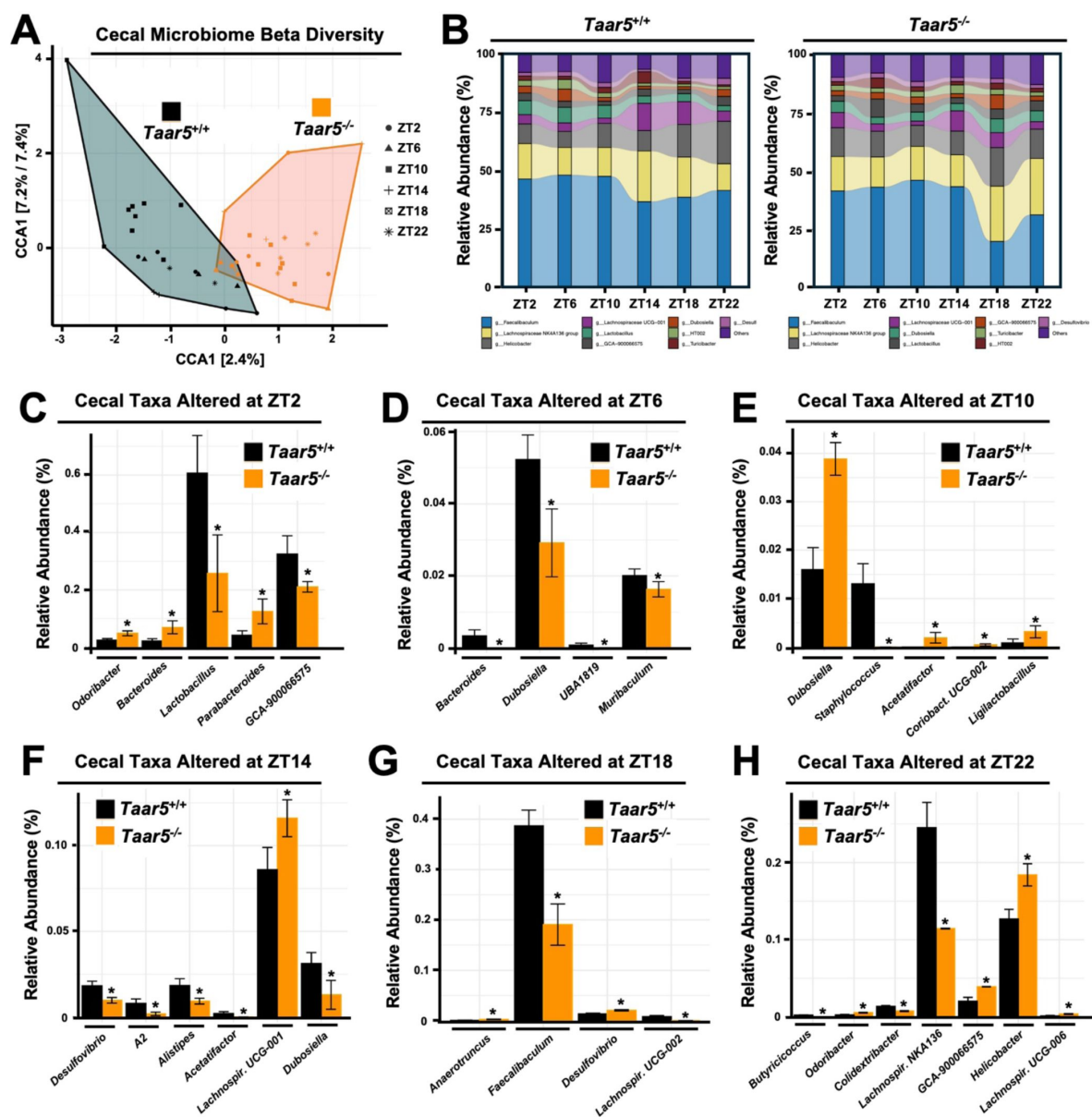


Figure 3.

The Trimethylamine Receptor TAAR5 Shapes Circadian Oscillations in the Gut Microbiome.

Male chow-fed wild type (*Taar5*^{+/+}) or mice lacking the TMA receptor (*Taar5*^{-/-}) were necropsied at 4-hour intervals to collect cecum for microbiome composition analyses via sequencing the V4 region of the 16S rRNA (genus level changes are shown).

(A) Canonical correspondence analysis (CCA) based beta diversity analyses show distinct microbiome compositions in *Taar5*^{+/+} and *Taar5*^{-/-} mice. Statistical significance and beta dispersion were estimated using PERMANOVA. (B) The relative abundance of cecal microbiota in *Taar5*^{+/+} and *Taar5*^{-/-} mice. Significantly altered cecal microbial genera in *Taar5*^{+/+} and *Taar5*^{-/-} mice are shown at ZT2 (C), ZT6 (D), ZT10 (E), ZT14 (F), ZT18 (G), and ZT22 (H). ASVs that were significantly different in abundance (MetagenomeSeq with Benjamini-Hochberg FDR multiple test correction, adjusted $p < 0.01$). Data shown represent the means $\pm$ S.D. for $n = 3$ -6 individual mice per group. Group differences were determined using ANOVA with Benjamini-Hochberg false discovery rate (FDR) multiple test correction, * = adjusted $p < 0.01$).

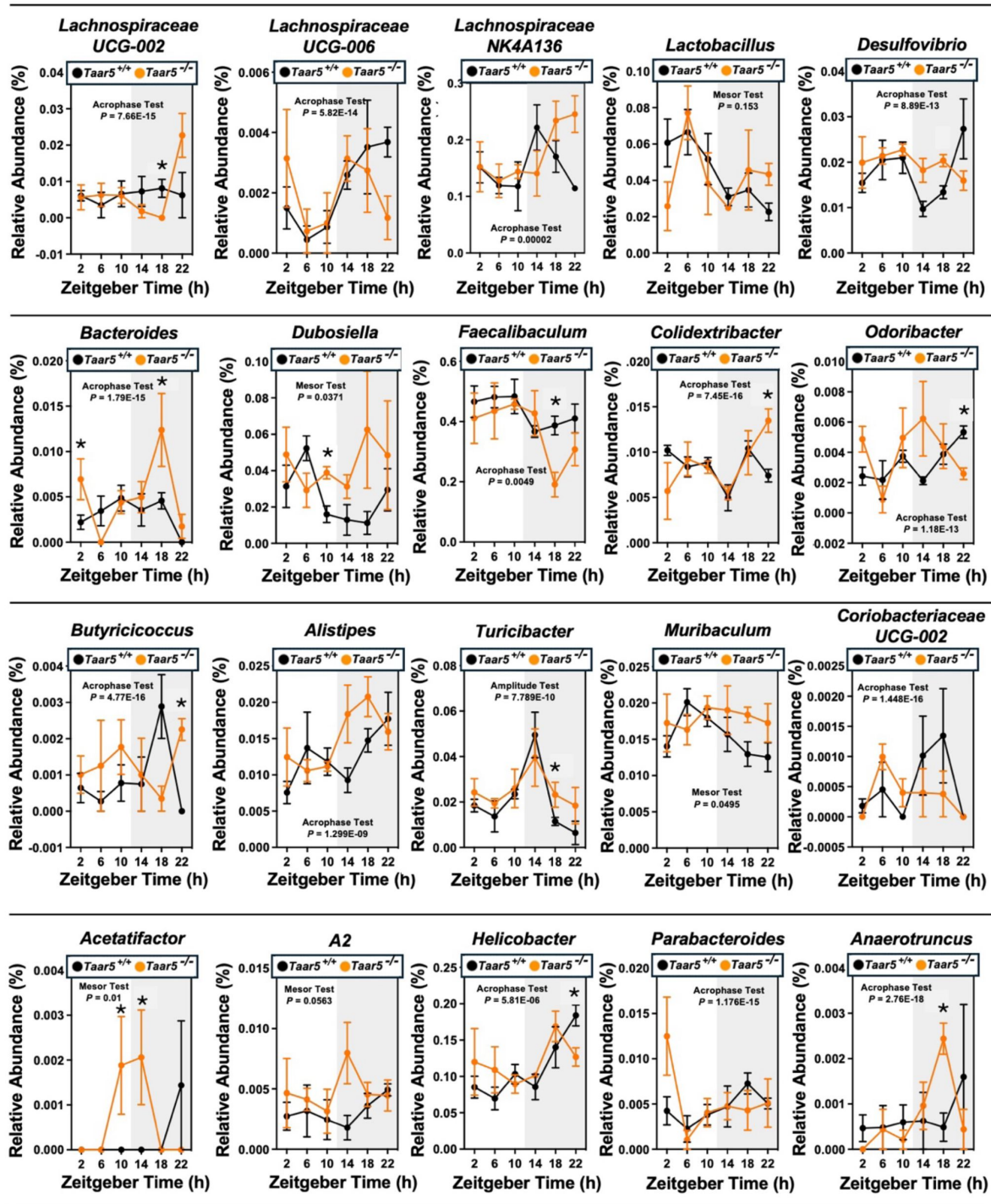


Figure 3 – Figure Supplement 1.

TAAR5-Deficient Mice Have Altered Circadian Oscillations in the Gut Microbiome.

Male chow-fed wild type ($Taar5^{+/+}$) or mice lacking the TMA receptor ($Taar5^{-/-}$) were necropsied at 4-hour intervals to collect cecum for microbiome composition analyses via sequencing the V4 region of the 16S rRNA (genus level changes are shown). The relative abundance of cecal microbiota at the genus level are shown, and group differences were determined using cosinor analyses p -values are provided where there were statistically significant differences between $Taar5^{+/+}$ and $Taar5^{-/-}$ mice. The complete cosinor statistical analysis for circadian data can be found in Table S1. * = Significant differences between $Taar5^{+/+}$ and $Taar5^{-/-}$ mice by student's t -tests within each ZT time point ($p < 0.05$).

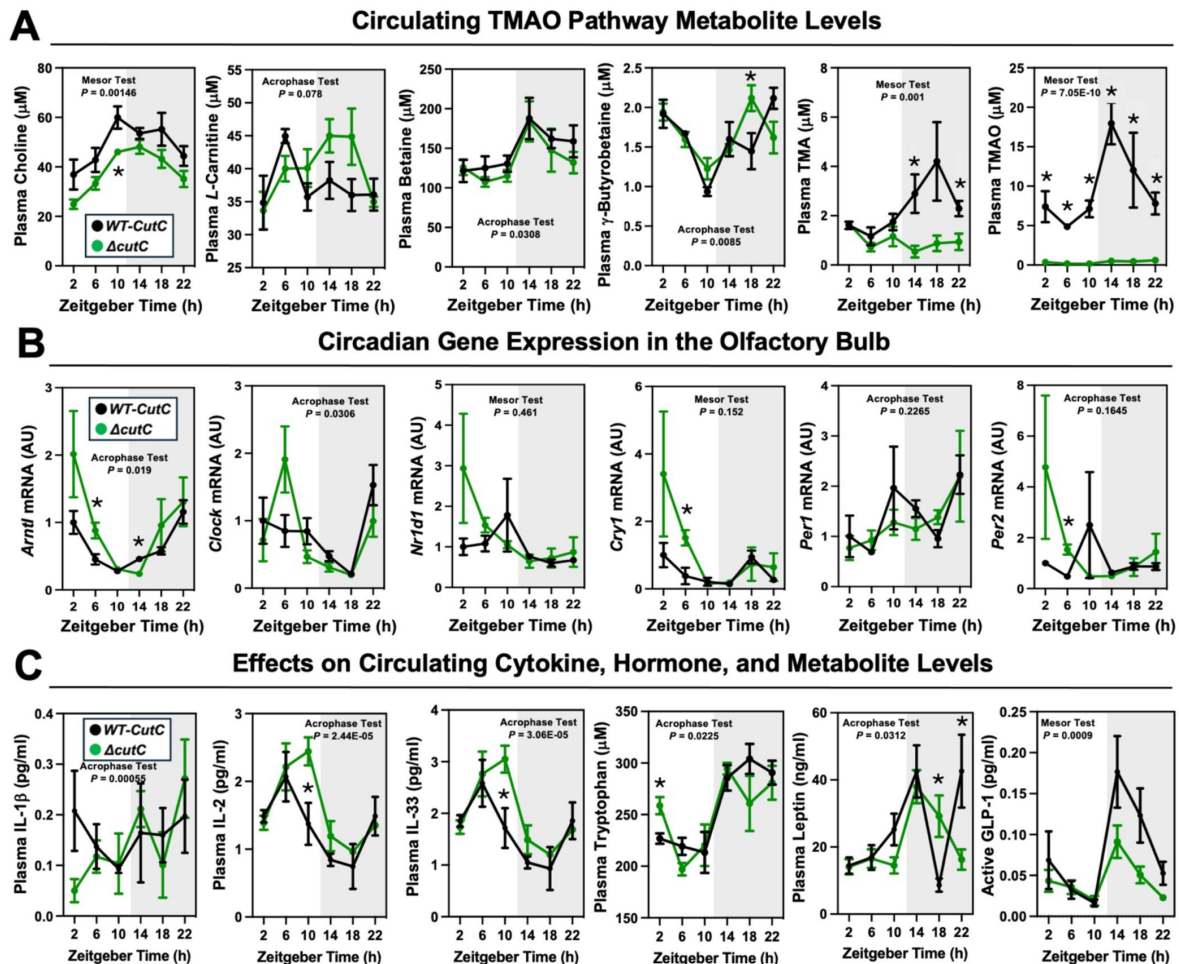


Figure 4.

Transplanting a Defined Synthetic Microbial Community With or Without Genetically Deleted Trimethylamine Production Capacity ($\Delta cutC$) Alters Host Circadian Rhythms.

Germ-free C57Bl/6 mice (recipients) were gavaged with the core community (*B. caccae*, *B. ovatus*, *B. thetaiotaomicron*, *C. aerofaciens*, *E. rectale*) with TMA producing wild type (WT) *C. sporogenes* (produces TMAO) or *C. sporogenes* $\Delta cutC$. Gnotobiotic mice were then necropsied at 4-hour intervals to collect tissues including plasma (**A** and **C**) and olfactory bulb (**B**). (**A**) Plasma choline, L-carnitine, betaine, ψ -butyrobetaine, trimethylamine (TMA), and trimethylamine N-oxide (TMAO) were quantified by liquid chromatography with online tandem mass spectrometry (LC-MS/MS). (**B**) PCR was performed on olfactory bulb to examine key circadian clock regulators. (**C**) Plasma levels of select cytokines including interleukins (IL-1b, IL-2, and IL-33) and other plasma metabolites and hormones were measured as described in the methods section. Data shown represent the means $\pm$ S.E.M. for $n = 5$ -6 individual mice per group. Differences between WT-*cutC* and $\Delta cutC$ groups were determined using cosinor analyses p -values are provided where there were statistically significant differences between group for circadian statistics. The complete cosinor statistical analysis for circadian data can be found in Table S1. Significant differences between WT-*cutC* and $\Delta cutC$ groups were also analyzed by student's t -tests within each ZT time point (* = $p < 0.05$ and ** = $p < 0.01$).

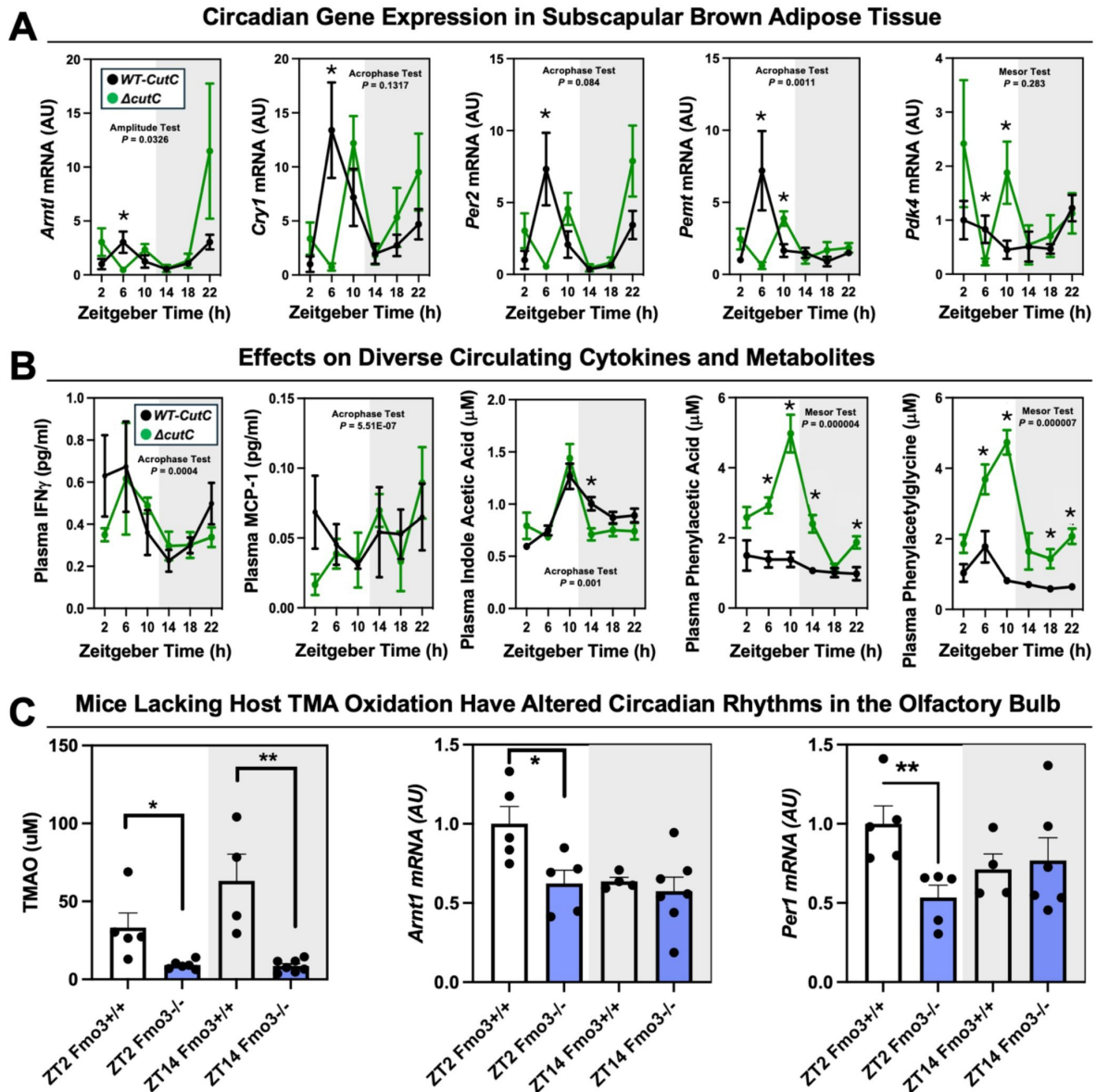


Figure 4 – Figure Supplement 1.

Mice Lacking Either Gut Microbial TMA Production or Host-Driven TMA Oxidation Have Altered Circadian Rhythms.

(A,B) Female germ-free C57Bl/6 mice (recipients) were gavaged with the core community (*B. caccae*, *B. ovatus*, *B. theta*, *taotomicon*, *C. aerofaciens*, *E. rectale*) with TMA producing wild type (WT) *C. sporogenes* (produces TMAO) or *C. sporogenes* $\Delta cutC$. Gnotobiotic mice were then necropsied at 4-hour intervals to collect tissues including subscapular brown adipose tissue **(A)** and plasma **(B)**. Gene expression was quantified by qPCR and cytokines and metabolite levels were quantified as described in the method section. Data for panels B and C were analyzed by cosinor analyses and representative p-values are shown; The complete statistical analysis for circadian metrics can be found in Table S1. **(C)** Female wild type (*Fmo3*^{+/+}) or flavin-containing monooxygenase 3 knockout (*Fmo3*^{-/-}) mice were necropsied at ZT2 or ZT14 and TMAO levels were measured by LC-MS/MS and olfactory bulb circadian gene expression was quantified by qPCR. Data shown represent the means $\pm$ S.E.M. for n= 4-6 individual mice per group. Group differences were determined using cosinor analyses p-values are provided where there were statistically significant differences between group for circadian statistics. The complete cosinor statistical analysis for circadian data can be found in Table S1. Significant differences between by student's t-tests within each ZT time point (* = $p < 0.05$ and ** = $p < 0.01$).

Discussion

The metaorganismal TMAO pathway is strongly associated with many human diseases^{5,29}, which has prompted the rapid development of drugs intended to lower circulating levels of TMAO.^{30–37} As the rapid drug discovery advances towards human studies, it will be extremely important to understand the diverse mechanisms by which the primary metabolite TMA and/or the secondary metabolite TMAO promotes disease pathogenesis in the human metaorganism. There is compelling evidence that the end product of the pathway TMAO can promote inflammation, ER stress and platelet activation via activation of NFκB, the NLRP3 inflammasome, PERK, and stimulus-dependent calcium release, respectively.^{28,38–41} However, these TMAO-driven mechanisms only partially explain the links to so many diverse human diseases. Here, we provide new evidence that in addition to these TMAO-driven mechanisms, the primary gut microbe TMA can in parallel shape circadian rhythms through the host GPCR TAAR5. The major findings of the current studies are: (1) Mice lacking the TMA receptor TAAR5 have abnormal oscillations in core circadian genes, particularly in the olfactory bulb, (2) Compared to wild type controls, *Taar5*^{−/−} mice have altered circulating levels of cytokines, metabolic hormones, and metabolites at certain times of the day, (3) *Taar5*^{−/−} mice have altered innate and repetitive behaviors that emerge only in a time of day-dependent manner, (4) The normal oscillatory behavior of the cecal microbiome is dysregulated in *Taar5*^{−/−} mice, (5) Genetic deletion of gut microbial choline TMA lyase activity, using defined *cutC*-null microbial communities *in vivo*, results in rewiring of circadian rhythms in gene expression, metabolic hormones, cytokines, and metabolites, and (6) Mice lacking the host liver TMA-to-TMAO converting enzyme FMO3 likewise have altered circadian gene expression in the olfactory bulb. Collectively, our findings suggest that therapeutic strategies designed to limit gut microbial TMA production (i.e. TMA lyase inhibitors), host liver TMA oxidation (i.e. FMO3 inhibitors), or sensing of TMA by the host GPCR TAAR5 (i.e. TAAR5 inhibitors) will need to be carefully evaluated for pleiotropic effects on circadian rhythms. This body of work also demonstrates that diet-microbe-host interactions can powerfully shape chronobiology, and provides one of the first examples of a microbial metabolite being sensed by a host GPCR to rewire circadian rhythms.

Currently, the only known host receptor that senses gut microbe-produced TMA is the volatile amine receptor TAAR5, which allows for species-specific recognition of the “fishy” odor intrinsic to TMA but not TMAO.^{47–49} In general, the trace amine-associated receptor (TAAR) subfamily of GPCRs primarily function as olfactory receptors in vertebrates.^{49,50} Ligand-dependent activation of each individual TAAR receptor occurs in a unique sensory neuron population primarily localized to olfactory cilia, and when activated typically elicit TAAR-specific olfactory-related behavior responses.^{49,50} TMA-dependent activation of TAAR5 was first shown to promote attraction and social interaction between females and male mice in a very species and strain-specific manner.⁴⁸ This seminal study by Liberles and colleagues showed that the TMA-TAAR5 olfactory circuit can powerfully shape olfactory-driven social interaction, and FMO3-driven conversion of TMA to TMAO suppresses TAAR5 activation.⁴⁸ In parallel, we recently performed a comprehensive study to assess cognitive, motor, anxiolytic, social, olfactory and innate behaviors in mice treated with small molecule choline TMA lyase inhibitors to block bacterial TMA production.⁴⁶ Much like the olfactory and social phenotypes seen in *Taar5*^{−/−} mice,⁴⁸ we found that pharmacologic blockade of bacterial TMA production significantly altered olfaction and olfactory-related social behaviors, but did not significantly alter cognitive, motor, anxiolytic behaviors.⁴⁶ It is interesting to note that choline TMA lyase inhibitors altered olfactory perception of several odorant stimuli beyond TMA itself including a cookie, almond, vanilla, corn oil, and coyote urine.⁴⁶ This indicates that drugs blocking TMA-TAAR5 signaling could impact olfactory neurogenesis to impact sensing of diverse odorant cues that are well beyond the rotten fish smell intrinsic to TMA. Given that anosmia is a common occurrence in several TMAO-related diseases including CKD,⁶¹ obesity, diabetes,⁶² neurodegenerative diseases,^{63,64} and

COVID-19⁶⁵, it is tempting to speculate that TMA-lowering therapeutics may hold promise to potential restore olfactory perception to diverse stimuli in these disease conditions. However, additional studies are needed to formally test this hypothesis. Given the clear links between the TMA-TAAR5 signaling axis and olfactory-related behaviors in mice^{46,48}, it will be important to carefully evaluate effects of TMAO-lowering drugs on olfaction in future human studies.

A commonality of many bacterially-derived volatile metabolites is that they have a pungent odor which is sensed by olfactory neurons, and these microbe-host olfactory circuits shape many diverse behavioral and metabolic responses in the human host. For example, strong bacterially-derived odors such as trimethylamine (i.e. smells like rotting fish), hydrogen sulfide (i.e. the rotten egg smell), polyamines including putrescine and cadaverine (i.e. the smell of rotting cadavers/roadkill) can serve as an effective “don’t eat me” signal for us to avoid consuming spoiled food. Within the human metaorganism, the gut microbiome and select gut microbe-derived metabolites play an important role in shaping not only food intake but also other aspects of host energy metabolism, insulin sensitivity, immunology, and the circadian clock^{56–60}. It is important to note that there is bi-directional communication between gut microbes and the host in maintaining the circadian rhythms in metabolism and immunity within mammalian metaorganisms^{56–60}. First, central and peripheral clock machinery in the host clearly shape circadian oscillations in the gut microbiome primarily through innate immune mechanisms^{56,57}. At the same time, circadian oscillations intrinsic to the gut microbiome itself can have profound impacts on host physiology and disease^{58–60}. Although there is extensive knowledge on how the host circadian clock can impact the oscillatory behavior of the gut microbiome^{56,57}, much less is known regarding mechanisms by which the gut microbiome can instruct the host circadian clock. Here, we provide one of the first examples of a gut microbe-derived metabolite signaling via a host GPCR to shape circadian rhythms in gene expression, metabolic hormones, gut microbiome composition, and innate behaviors. Using both pharmacologic³³ and genetic approaches, we have demonstrated that blocking either gut microbial TMA production or sensing by the host TMA receptor TAAR5 results in striking rearrangement of circadian rhythms in the gut microbiome that are linked to metabolic homeostasis and olfactory-related behaviors in mice^{33,46}. In further support of the connections between the TMAO pathway and host circadian disruption, a recent paper showed that FMO3 inhibitors blocking host TMA oxidation improved learning and memory deficits driven by chronic sleep loss⁶⁶. Other recent studies in humans also show that TMAO is elevated with sleep deprivation⁶⁷, and elevated TMAO levels are associated with an “evening chronotype” that is linked to increased risk of cardiometabolic disease⁶⁸. Collectively, the metaorganismal TMAO pathway plays an important in integrating dietary cues and microbe-host interactions in circadian rhythms. Given TMAO lowering drugs are rapidly advancing towards human studies, it will be of great interest to determine whether these drugs can improve circadian rhythms and potentially synergize with other developmental chronotherapies to improve human health.

Limitations

First, it is critically important to point out that the TMA-TAAR5 signaling axis has evolved to elicit species-specific behavioral responses that differ quite substantially across rodent species and humans^{48,49}. When the “stinky” fish odor of TMA is sensed by rat or human TAAR5, the result is typically aversion from consuming what is perceived as rotting food^{48,49}. However, select strains of mice have concentration-dependent attraction to low levels of TMA or aversion to high levels of TMA⁴⁸. It has been theorized that this concentration-dependent behavioral response to TMA in mice may have evolved as a kairomone signal that allows mice to sense approaching prey animals^{48,49}. Clearly the human TMA-TAAR5 signaling circuit evolved for different reasons in humans, where predator-prey signaling circuits are less prominent. Therefore, additional studies are needed to better understand what diverse roles TMA-driven TAAR5 activation plays in the circadian rhythms in human metabolism and behavior. The recent discovery of humans with loss-of-function TAAR5 genetic variants⁶⁹ may provide a unique population to confirm and extend the findings we report here in mice. Another limitation in our current understanding is that drugs

blocking gut microbial TMA production^{30–37} or host TMA oxidation^{13,15,66} have only been tested in mice. Although these drugs show clear benefit in mice models, the human gut microbiome is quite different than mice and the relative affinity of TMA for TAAR5 is ~ 200-fold less in humans when compared to rodent forms of the receptor^{47–49}. Therefore, it is impossible to extrapolate finding in mice to humans. As TMAO lowering drugs move into human trials, there is strong potential that human-specific phenotypes may emerge. A final limitation here is that here we have studied mice globally lacking TAAR5 in all tissues and cell types. Although TAAR5 has primarily been studied in olfactory neurons in the brain, TAAR5 is also expressed in immune cells⁷⁰ as well as skeletal and cardiac muscle³³. Given the expression of TAAR5 outside of the central nervous system, future studies utilizing cell-restricted *Taar5* knockout mice will provide clues into the cell autonomous roles for TMA-TAAR5 signaling. Despite these limitations, a growing body of evidence links TMA-driven TAAR5 activation to circadian regulation of metabolism and olfactory-related behaviors.

STAR★Methods

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
<i>Bacteroides thetaiotaomicron</i>	ATCC	VPI-5482
<i>Bacteroides caccae</i>	ATCC	ATCC 43185
<i>Bacteroides ovatus</i>	ATCC	ATCC 8483
<i>Collinsella aerofaciens</i>	ATCC	ATCC 25986
<i>Eubacterium rectale</i>	ATCC	ATCC 33656
<i>Clostridium sporogenes</i>	ATCC	ATCC 15579
<i>Clostridium sporogenes</i> $\Delta cutC$	Dr. Michael Fischbach at Stanford University	N/A
Chemicals, peptides, and recombinant proteins		
Choline chloride	Sigma	#C7017
Choline chloride (trimethyl-D ₉ , 98%)	Cambridge Isotope Laboratories, Inc.	#DLM-549-MPT-PK
Betaine	Sigma	#61962
N-(Carboxymethyl)-N,N,N-trimethyl-d ₉ -ammonium Chloride	C/D/N Isotopes Inc.	#D-3352
L-Carnitine hydrochloride	Sigma	#94954
L-Carnitine-d ₃ HCl (N-methyl-d ₃)	C/D/N Isotopes Inc.	#D-5069
(3-Carboxypropyl)trimethylammonium chloride (butyrobetaine)	Sigma	#403245
Butyrobetaine-d ₉	Wang, Z. <i>et al.</i> ³⁰	NA
5-hydroxyindole-3-acetic acid	Sigma	#H8876
5-Hydroxyindole-3-Acetic-2,2-D ₂ Acid	CDN Isotopes	#D-1547
hippuric acid	Sigma	#8206490100
N-Benzoyl-d ₅ -glycine	CDN Isotopes	#D-5588
4-OH-Hippuric acid	Santa Cruz Biotechnology	#SC-277427
3-hydroxyhippuric acid	TRC	#H943125
2-Hydroxyhippuric Acid	Carbosynth	#FH240191601
Indole-3-Propionic Acid	Alfa Aesar	#L04877
Indole-3-Propionic-2,2-D ₂ Acid	CDN Isotopes	#D-7686
Methylindole-3-Acetate	Sigma	#19770
DL-Indole-3-Lactic Acid	Chem-Impex	#21729
Phenylacetic acid	Aldrich	#P16621
Phenylacetic Acid-1,2- ¹³ C ₂	Chem Cruz	#SC-236371
DL-p-Hydrophenyllactic Acid	Aldrich	#H3253
Indoxyl-Glucuronide	Abcam	#Ab146380
L-Tryptophan-2',4',5',6',7'-d ₅ (indole-d ₅)	CDN Isotopes	#D-1522
serotonin	Sigma	#H-9523
Tryptamine	Aldrich	#193747
Tryptamine- $\alpha,\alpha,\beta,\beta$ -D ₄ HCl	CDN Isotopes	#D-1546
3-Indoleacetic Acid	Aldrich	#13750
Indole-3-Acetic-2,2-D ₂ Acid	CDN Isotopes	#D-1709
Sodium phenyl sulfate	Enamine	#EN300-1704008
TMAO	Sigma	#317594
Trimethylamine N-oxide (D ₉ , 98%)	Cambridge Isotope Laboratories, Inc.	#DLM-4779-1

Phenylacetyl-L-Glutamine	Chem-Impex	#16414
N-alpha-(phenyl-d5-acetyl)-L-glutamine	CDN Isotopes	#D-6900
Potassium p-tolyl sulfate P-cresol sulfate K-salt)	TCI	#P2091
p-Cresol sulfate, potassium salt (D ₇ , 98%)	Cambridge Isotope Laboratories, Inc.	#DLM-9786
3-Indoxyl Sulfate Potassium Salt	Chem-Impex	#21710
3-Indoxyl Sulfate-d4 Potassium Salt	TRC	#I655102
Phenylacetylglycine	Bachem	#4016439
trans-3-Indoleacrylic acid	Aldrich	#I3807
4-Ethylphenyl sulfate Potassium salt	TRC	#E925865
Trimethylamine hydrochloride	Sigma	#41284
Trimethylamine:DCI (D ₁₀ , 98%)	Cambridge Isotope Laboratories, Inc.	#DLM-1817-5
Critical commercial assays		
Monarch Total RNA Miniprep Kit	New England BioLabs	#T2010S
V-Plex Cytokine Panel 1 Mouse Kit	Mesoscale Discovery	#K15245D
V-Plex Pro-Inflammatory Panel 1 Mouse Kit	Mesoscale Discovery	#K15048D
U-Plex Metabolic Combo Mouse Kit	Mesoscale Discovery	#K15297K
Ultra Sensitive Mouse Insulin ELISA	Crystal Chem Inc.	#90080
DNeasy PowerSoil Pro Kit	Qiagen	#47014
Experimental models: Organisms/strains		
Germ/free C57BL/6NTac	Taconic	Stock #: B6-GF-F
C57BL/6J	Jackson	#00664
C57BL/6J <i>Taar5</i> ^{-/-}	This paper	NA
C57BL/6J <i>Fmo3</i> ^{-/-}	Massey <i>et al.</i> ⁷³	NA
Oligonucleotides		
<i>LacZ</i> Forward: CCAACGTGACCTATCCCATTAC	Sigma	NA
<i>LacZ</i> Reverse: ATCTTCCTGAGCCGATACT	Sigma	NA
<i>Arntl1</i> Forward: CCAAGAAAGTATGGACACAGACAAA	Sigma	NA
<i>Arntl1</i> Reverse: GCATTCTTGATCCTTCCTTGCT	Sigma	NA
<i>Nr1d1</i> Forward: ATGCCAATCATGCATCAGGT	Sigma	NA
<i>Nr1d1</i> Reverse: CCCATTGCTGTTAGGTTGGT	Sigma	NA
<i>Per1</i> Forward: TGTCTGGTTTCGAAGTGTG	Sigma	NA
<i>Per1</i> Reverse: TGTGTCAAGCAGGTTTCAGG	Sigma	NA
<i>Per2</i> Forward: GCTGACGCACACAAAGAACT	Sigma	NA
<i>Per2</i> Reverse: TAGCCTTCACCTGCTTCACG	Sigma	NA
<i>Clock</i> Forward: AGGCACAGACATTATCGG	Sigma	NA
<i>Clock</i> Reverse: ACCGTCTCATCAAGGGAC	Sigma	NA
<i>Cry1</i> Forward: TACTGGGAAACGCTGAACCC	Sigma	NA
<i>Cry1</i> Reverse: ACCCCAAGCTTGTTCCTAA	Sigma	NA
<i>Cry2</i> Forward: GCTGGAAGCAGCCGAGGAACC	Sigma	NA
<i>Cry2</i> Reverse: GGGCTTTGCTCAGGAGCGA	Sigma	NA
<i>Prdm16</i> Forward: CAGCACGGTGAAGCCATTC	Sigma	NA
<i>Prdm16</i> Reverse: GCGTGCATCCGCTTGTG	Sigma	NA
<i>Ucp1</i> Forward: ACTGCCACACCTCCAGTCATT	Sigma	NA
<i>Ucp1</i> Reverse: CTTTGCCTCACTCAGGATTGG	Sigma	NA
<i>Pemt</i> Forward: TGTGCTGTCCAGCTTCTATG	Sigma	NA
<i>Pemt</i> Reverse: GAAGGGAAATGTGGTCACTCT	Sigma	NA
<i>Pdk4</i> Forward: GTGCTCTCTGGTCTCTGTG	Sigma	NA
<i>Pdk4</i> Reverse: AGTCCAACGGACAAAACGGA	Sigma	NA
<i>Fmo3</i> Forward: CCCACATGCTTTGAGAGGAG	Sigma	NA

<i>Fmo3</i> Reverse: GGAAGAGTTGGTGAAGACCG	Sigma	NA
<i>CycloA</i> Forward: GCGGCAGGTCCATCTACG	Sigma	NA
<i>CycloA</i> Reverse: GCCATCCAGCCATTCTAGTC	Sigma	NA
<i>Gapdh</i> Forward: CCTCGTCCCGTAGACAAAATG	Sigma	NA
<i>Gapdh</i> Reverse: TGAAGGGGTCGTTGATGGC	Sigma	NA
Software and algorithms		
GraphPad Prism	https://www.graphpad.com/	Version 10.4.1
DADA2	https://benjjneb.github.io/dada2/	Version 3.16
metagenomeSeq	https://github.com/HCBravoLab/metagenomeSeq	Version: Release (3.20)
DAtest	https://github.com/Rusel88/DAtest	Version 2.8.0.
ggplot2	https://cran.r-project.org/web/packages/ggplot2/index.html	Version 3.5.1
cosinor	https://cran.r-project.org/web/packages/cosinor/index.html	Version 1.2.3
cosinor2	https://cran.r-project.org/web/packages/cosinor2/index.html	Version 0.2.1
EthoVision XT15 (video tracking software)	Noldus	NA
Other		
qScript	QuantaBio	#95048-100
Fast SYBR Green Master Mix	Applied Biosystems	#4385612
Nutter Butter Cookies	Nabisco	NA
Marbles	Moon Marble Company	NA
SR-LAB-Startle Response System	San Diego Instruments	#2325-0400
Elevated Plus Maze	Nationwide Plastics	NA
Y Maze	Nationwide Plastics	NA
Forepaw Grip Strength Meter	Columbus Instruments	#1027SM
Rotamex-5	Columbus Instruments	#0254-8000
Fear Conditioning Chambers	Med Associates	#VFC-008
Hot Plate	Ugo Basile	#35300
OxyMax Clams Home Cage System	Columbus Instruments	NA

Experimental procedures

Animal Studies (Mice)

Taar5^{-/-} mice [*Taar5*^{tm1.1}(KOMP)^{Vlcg}] used here were created from ES cell clone 10675A-A8 and originated on the C57BL/6N background by Regeneron Pharmaceuticals, Inc. The successfully targeted ES cells were then used to create live mice by the KOMP Repository and the Mouse Biology Program at the University of California Davis. It is important to note that this KO/reporter mouse line has the removal of the neomycin selection cassette and critical exon(s) leaving behind the inserted *lacZ* reporter sequence knocked into the *Taar5* locus. Given the well appreciated differences in metabolic homeostasis between C57BL/6N and C57BL/6J substrains^{71,72}, we backcrossed this line > 10 generations into the C57BL/6J congenic strain and backcrossing was confirmed by mouse genome SNP scanning at the Jackson Laboratory (Bar Harbor, ME). Heterozygous mice on a pure C57BL/6J background were bred to generate littermate *Taar5*^{+/+} and *Taar5*^{-/-} mice. Global *Fmo3*^{-/-} mice on a pure C57BL/6J background have been previously described⁷³. All mice were fed standardized control chow diet with defined sufficient level of dietary choline (Envigo diet TD:130104) that was twice irradiated for sterilization. For studies of

circadian rhythms, 9-week-old C57Bl6/J male mice were adapted for 2 weeks with a strict 12:12 light:dark cycle. After 7 days, plasma and tissue were collected every 4 hours over a 24-hour period. All dark cycle necropsies were performed under red light conditions to avoid light-driven circadian cues. For the CutC mutant studies shown in main **Figure 4**, 7-8 week-old germ-free B6/N mice were purchased from Taconic and placed on a sterile chow. Two weeks later, the mice were given an oral gavage of our 6 member community that included *B. caccae*, *B. ovatus*, *B. theta*, *C. aerofaciens*, *E. rectale*, and either wildtype or mutant *C. sporogenes* suspended in sterile PBS + 20% glycerol as previously described.⁵⁵ Gnotobiotic colonized mice were maintained using the Allentown Sentry SPP Cage System (Allentown, NJ) on a 14 hr:10 hr light:dark cycle. Mice were transferred to a room equipped with 12:12 light-dark cycling while still housed on Allentown Transport Carts. After one week of acclimation, plasma and tissue collection were performed every 4 hours over a 24-hour period. All dark cycle necropsies were performed under red light conditions. For all studies plasma was collected by cardiac puncture, and liver, GWAT, BAT and olfactory bulb were collected, flash-frozen, and stored at -80°C until the time of analysis. All mice were maintained in an Association for the Assessment and Accreditation of Laboratory Animal Care, International-approved animal facility, and all experimental protocols were approved by the Institutional Animal Care and use Committee of the Cleveland Clinic (Approved IACUC protocol numbers 00001941, 00002609, and 0002866).

RNA and Realtime PCR Methods

RNA isolation and quantitative polymerase chain reaction (qPCR) was conducted using methods previously described with minor modifications.^{13,33,46} To extract RNA, the Monarch Total RNA MiniPrep Kit (New England BioLabs, Inc.) was employed, and 325 ng of cDNA was utilized for Realtime PCR on a 384-well plate, using Applied Biosystem's Power SYBR Green PC mix in Roche LightCycler 480 I1. The levels of induced mRNAs were standardized by the $\Delta\Delta\text{CT}$ method and were normalized to cyclophilin A. The specificity of the primers was verified by analyzing the melting curves of the PCR products. Primer information is available in the Star Methods table.

Quantification of Plasma Metabolites Using Liquid Chromatography Tandem Mass Spectrometry (LC-MS/MS)

Stable isotope dilution high-performance LC-MS/MS was used for quantification of levels of TMAO, TMA, choline, carnitine, and ψ -butyrobetaine and all other reported circulating plasma metabolites as previously described performed on a Shimadzu 8050 triple quadrupole mass spectrometer.^{13,33,46} A subset of samples were also analyzed on a Thermo Vanquish liquid chromatograph coupled with a Thermo TSQ Quantiva mass spectrometer. 0.2% of formic acid in water (LC-MS grade) was used as mobile phase A and 0.2% formic acid in acetonitrile was used as mobile phase B. The separation was conducted using the following gradient: 0 min, 5% B; 0-2 min, 5% B; 2-8 min, 5%-100%B; 8-16 mins, 100%B; 16-16.5 mins, 5%B; 16.5-25 mins, 5%B. The flow rate was set at 0.2 ml/min. Samples were injected at 2 μl onto a Phenomenex Gemini C18 analytical column (2.0 $\times$ 150 mm, 3 μm). Column temperature was set at 25°C . The acquired data were processed by Thermo Xcalibur 4.3 software to calculate the concentrations. Both methods used multiple reaction monitoring of precursor and characteristic products as we have previously described.^{13,33,46}

Plasma Hormone and Cytokine Quantification

Plasma hormones and cytokine levels were quantified using U-PLEX (product # K15297K) and V-PLEX (K15297K) assays per the manufacturer's instructions (Meso Scale Diagnostics, Rockville, Maryland, USA).

Olfactory Cookie Test

To broadly understand olfactory perception, we performed the olfactory cookie test using methods previously described.⁴⁶ Briefly, one day prior to testing, food was removed from the home cage. The following day, a clean cage was filled with ~1.5 inches of clean bedding and a small portion of peanut butter cookie (Nutter Butter, Nabisco®) was placed randomly ~1 cm under the bedding. A test mouse was placed into the cage and given 10 minutes to find the hidden cookie. Latency to find the cookie was recorded.

Marble Burying Test

To test for innate/repetitive behavioral alterations, mice underwent a marble burying test. A clean mouse cage was filled with 2.5-3 inches of bedding, 20 black marbles were placed evenly in 5 rows of 4 across the bedding. A test mouse was placed into the cage and was allowed to bury marbles for 30 minutes. After 30 minutes the number of marbles buried was counted. A marble was defined as buried when < 25% of the marble was visible.

Pre-Pulse Inhibition and Startle Test

Mice underwent a PPI and startle threshold test using methods previously described.⁴⁶ On day 1 PPI was measured and on Day 2 startle threshold was measured. Mice were placed inside holding cylinders that sit atop of a piezoelectric accelerometer that detects and transduces animal movement (SR-LAB Startle Response System; SD Instruments; San Diego, CA). Acoustic stimuli were delivered by high-frequency speakers mounted 33 cm above the cylinders. Mouse movements were digitized and stored using computer software supplied by San Diego Instruments. From the onset of startle stimuli, 65 1-ms readings were recorded, and the amplitude of the startle responses was obtained in arbitrary units. Chambers were calibrated before testing, and sound levels were monitored using a sound meter (Tandy). For prepulse inhibition (PPI), mice were subjected to five trial types in a 22 min session: pulse alone (40 ms, 120 dB, white noise pulse), three different prepulse/pulse trials (20 ms prepulse of 4, 8, or 16 dB above background noise level of 70 dB preceded the 120-dB pulse by 100 ms onset-onset interval), and no stimulus. All trials were presented pseudo randomly with an average of 15 s (7–23 s) between the 62 trials. Testing began with a 5 min acclimation to the cylinders, followed by four blocks of test trials. The first and last blocks consisted of six pulse-alone trials. Blocks 2 and 3 each contained six pulse-alone trials, five of each level of prepulse/pulse trials, and five no-stimulus trials. Data were analyzed for baseline startle amplitude (initial pulse-alone trials) and PPI (percentage decrease in startle amplitude for prepulse/pulse trials compared with pulse-alone trials). The next day the startle reactivity test was conducted in a similar manner except eight presentations of six trial types were given in pseudorandom order: no stimulus, 80, 90-, 100-, 110-, or 120-dB pulses. Mean startle amplitudes for each condition were calculated.

Forepaw Grip Strength Test

Grip strength was measured using a digital grip meter (Chatillon Gauge with Mesh bar; Columbus Instruments, Columbus, OH) in which a mouse was suspended by its tail and allowed to grab onto a stainless-steel grid bar that is attached to a force transducer. The mouse was gently pulled away from the meter in a horizontal plane. The force applied to the bar immediately before the mouse released its grip was recorded in grams force by the digital sensor. Mice completed 10 trials with an intertrial interval of at least 30 seconds. The mouse's reported grip strength was the average of all 10 trials.

Hotplate Sensitivity Test

For the hotplate sensitivity test, each mouse received a single trial that lasted a maximum of 30 seconds, and the behavioral response was scored live. To assess sensitivity to hot stimuli, a hot plate (Ugo Basile, Italy) was warmed to 52°C and a mouse was placed on the plate and a tall plexiglass cylinder was placed around the mouse to prevent the animal from leaving. A foot pedal was pressed to start the timer on the hotplate. As soon as the mouse showed a response to the stimuli such as jumping or licking their paws, the foot pedal was pressed again to stop the timer and the mouse was removed from the plate. Latency to respond was recorded.

Rotarod

This was a 2-day task in which mice were placed onto a nonrotating rod (3 cm in diameter) that had ridges for a mouse to grip and was divided into 4 sections that allowed 4 mice to be tested at the same time (Rotamex-5; Columbus Instruments; Columbus, OH). Each mouse was subjected to 4 trials per day with an inter-trial interval of at least 30 minutes to prevent its performance from being impaired by fatigue. For each trial, a mouse was placed onto the rod and allowed to balance itself. When mice were balanced on the rod, the rod started rotating and accelerated from 4 to 40 rpm over a 5-minute period. Latency to stay onto the rod was assessed with the use of an infrared beam just above the rod and was broken once the mouse fell off the rod. Each mouse was subjected to a total of 8 trials over a 2-day period (4 trial/day). Latency to fall (seconds) was recorded.

Social Behavior

Social interaction with a juvenile was performed in a novel empty home cage under dim white light as described previously.⁴⁶ Briefly, following a 15-min habituation, a test mouse was placed in a novel empty cage with a juvenile stimulus mouse (BALB/cJ juvenile mouse; 3 weeks of age; Stock #000651 Jackson Labs) and was allowed to directly interact for 2 min. Interaction was scored by observing the duration and number of times the test mouse-initiated contact with or sniffed the juvenile mouse. Contact was considered as any part of the body touching the other mouse. Three days later the same test mouse and juvenile mouse were paired again in a novel clean cage for 2 min and scored in the same manner.

Fear Conditioning

Fear conditioning (FC) consisted of a training period and a testing period. For the training period, mice were placed in the fear conditioning boxes (Med Associates; Fairfax VT), with a gridded floor, an inside white light and NIR light, and a fixed NIR camera for video recording. Mice were placed inside the chamber for 2 minutes, and then a 30 second, 90 dB acoustic conditioned stimulus (CS; white noise) co-terminated with a 2 second 0.6 mA foot shock (US). Mice received a total of 3 US-CS pairing, with each pairing separated by 1 minute. The mouse remained in the chamber for 30 seconds after the pairings before returning to its home cage. The testing period occurred 24-hrs after the training period ended. The mouse's freezing behavior (motionless except for respirations) was monitored during testing to access memory. Contextual FC consisted of placing the mouse back into the same chamber as was done with training for 5 minutes and measuring freezing behavior. No shock or sound was presented to the mouse, just the context of the chamber. Later that day, mice were brought back to test for cue fear conditioning. Prior to cue FC testing, the inside of the FC chamber was modified using white plastic sheets, interior lighting, and vanilla extract on a paper towel placed in the bedding chamber to make the chamber look and smell differently than the training period. Mice were placed into this modified chamber and allowed to habituate for 3 minutes. Following 3 minutes the same white noise tone used in training was presented to the mouse for 3 minutes. During the tone presentation mouse freezing behavior was recorded. Total percent freezing for context and cue FC was recorded.

Elevated Plus Maze

The elevated plus maze task was conducted essentially as described previously.⁶⁴ Mice were placed in the center of a black, plexiglass elevated plus maze (each arm 33 cm long and 5 cm wide with 25 cm high walls on closed arms) in a dimly lit room for 5 minutes. Automated video tracking software from Noldus Ethovision XT13 (Leesburg VA) was used to track time spent in the open and closed arms, number of open and closed arm entries, and number of explorations of the open arm (defined as placing head and two limbs into open arm without full entry).

Y-Maze

Spontaneous alteration in Y-Maze was assessed as previously described.⁶⁴ Briefly, a mouse was placed into one arm of the Y-maze facing the center (14"). Video-tracking was used to record the spontaneous behavior of each mouse for 10 min. Zone (i.e. arm) alteration was later analyzed using Noldus EthoVision XT14, to determine when a mouse entered 3 consecutive different arms of the maze. Spontaneous alteration % was calculated as follows: Alteration % = #spontaneous alterations/ total number of arm entries - 2 x 100.

Indirect Calorimetry and Metabolic Cage Measurements During Cold Challenge

Cold-induced metabolic responses were quantified using the Oxymax CLAMS system (Columbus Instruments) as previously described.¹³ Briefly, weight matched *Taar5*^{-/-} and *Taar5*^{+/+} mice were acclimated to metabolic cages for 48 hours. Thereafter, physical activity, oxygen consumption (VO₂), carbon dioxide production (VCO₂), and respiratory exchange ratio (RER) were continually monitored for 24 hours at thermoneutrality (30°C), 24 hours at room temperature 22°C, and 24 hours in the cold (4°C). Data represent the last 6 a.m. to 6 a.m. period after adequate acclimation.

16S rRNA gene amplicon sequencing and bioinformatics

16S rRNA gene amplicon sequencing and bioinformatics analysis were performed using our published methods.³³ Briefly, raw 16S amplicon sequence and metadata, were *demultiplexed* using *split_libraries_fastq.py* script implemented in *QIIME2*.⁷⁴ Demultiplexed fastq file was split into sample-specific fastq files using *split_sequence_file_on_sample_ids.py* script from *QIIME2*. Individual fastq files without non-biological nucleotides were processed using Divisive Amplicon Denoising Algorithm (DADA) pipeline.⁷⁵ The output of the dada2 pipeline [feature table of amplicon sequence variants (an ASV table)] was processed for alpha and beta diversity analysis using *phyloseq*.⁷⁶, and *microbiomeSeq* (<http://www.github.com/umerijaz/microbiomeSeq>) packages in R. We analyzed variance (ANOVA) among sample categories while measuring the of α -diversity measures using *plot_anova_diversity* function in *microbiomeSeq* package. Permutational multivariate analysis of variance (PERMANOVA) with 999 permutations was performed on all principal coordinates obtained during CCA with the *ordination* function of the *microbiomeSeq* package. Pairwise correlation was performed between the microbiome (genera) and metabolomics (metabolites) data was performed using the *microbiomeSeq* package.

16S Statistical Analysis

Differential abundance analysis was performed using the random-forest algorithm, implemented in the *DAtest* package (<https://github.com/Russel88/DAtest/wiki/usage#typical-workflow>).

Briefly, differentially abundant methods were compared with False Discovery Rate (FDR), Area Under the (Receiver Operator) Curve (AUC), Empirical power (Power), and False Positive Rate (FPR). Based on the *DAtest*'s benchmarking, we selected *lefseq* and *anova* as the methods of choice to perform differential abundance analysis. We assessed the statistical significance ($P < 0.05$)

throughout, and whenever necessary, we adjusted *P*-values for multiple comparisons according to the Benjamini and Hochberg method to control False Discovery Rate.⁷⁷ Linear regression (parametric test), and Wilcoxon (Non-parametric) test were performed on genera and ASVs abundances against metadata variables using their base functions in R (version 4.1.2; R Core Team, 2021).

Data Analyses for Circadian Rhythmicity (Cosinor Analyses)

A single cosinor analysis was performed as previously described.^{78,79} Briefly, a cosinor analysis was performed on each sample using the equation for cosinor fit as follows:

$$Y(t)=M+A\cos(2\pi t/\tau+\Phi)$$

where *M* is the MESOR (midline statistic of rhythm, a rhythm adjusted mean), *A* is the amplitude (a measure of half the extent of the variation within the cycle), Φ is the acrophase (a measure of the time of overall highest value), and τ is the period. The fit of the model was determined by the residuals of the fitted wave. After a single cosinor fit for all samples, linearized parameters were then averaged across all samples allowing for calculation of delinearized parameters for the population mean. A 24 hr period was used for all analysis. Comparison of population MESOR, amplitude, and acrophase was performed as previously described.³³ Comparisons are based on *F*-ratios with degrees of freedom representing the number of populations and total number of subjects. All analyses were done in R v.4.0.2 using the cosinor and cosinor2 packages.^{79–81}

Standard Statistical Analyses

Data are expressed as the mean $\pm$ standard error of the mean (SEM). All data were analyzed using either one-way or two-way analysis of variance followed by Student's *t* tests for post hoc analysis. Differences were considered significant at *p* < 0.05. All analyses were performed using GraphPad Prism software (version 10.2.2; GraphPad Software, Inc).

Data and code availability

All multi-omics datasets are currently being submitted to publicly-available data portals, and accession numbers will be provided upon acceptance of this work.

Code used for cosinor analyses: Sachs (2014)⁷⁸. Any additional information required to reanalyze the data reported in this paper is available from the lead contact (Dr. J. Mark Brown) upon request

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	Gene	Group	Zero Amplitude Test	MESOR	Amplitude	Acrophase	Mesor Test	Amplitude Test	Acrophase Test
Figure 1A: Skeletal Muscle Gene Expression	<i>LacZ</i>	Taar5 +/+	1.82E-02	4.754 ± 24.113	9.899 ± 33.683	0.601 ± 6.688	2.85E-04	1.82E-02	2.58E-01
		Taar5 -/-	1.19E-01	141.511 ± 34.834	251.811 ± 35.373	-1.23 ± 0.327			
	<i>Amt1</i>	Taar5 +/+	6.53E-02	0.555 ± 0.062	0.796 ± 0.086	0.82 ± 0.368	2.58E-01	8.30E-01	1.61E-01
		Taar5 -/-	2.84E-02	0.497 ± 0.089	0.735 ± 0.098	0.981 ± 0.36			
	<i>Nr1d1</i>	Taar5 +/+	1.24E-07	0.553 ± 0.048	1.072 ± 0.066	1.126 ± 0.132	3.79E-01	6.28E-01	2.76E-03
		Taar5 -/-	6.09E-05	0.521 ± 0.069	0.961 ± 0.074	1.445 ± 0.153			
	<i>Per1</i>	Taar5 +/+	6.67E-05	4.129 ± 0.987	7.785 ± 1.37	0.954 ± 0.388	3.38E-01	6.53E-01	2.10E-01
		Taar5 -/-	5.31E-02	5.215 ± 1.426	10.66 ± 1.547	0.819 ± 0.252			
	<i>Per2</i>	Taar5 +/+	1.45E-04	2.679 ± 0.198	4.415 ± 0.278	0.347 ± 0.161	9.94E-01	6.54E-01	1.03E-01
		Taar5 -/-	2.26E-07	2.708 ± 0.286	4.364 ± 0.285	0.05 ± 0.182			
Figure 1B: Olfactory Bulb Gene Expression	<i>Amt1</i>	Taar5 +/+	1.26E-01	1.167 ± 0.086	1.408 ± 0.117	1.155 ± 0.526	2.44E-03	6.28E-01	8.57E-02
		Taar5 -/-	1.12E-02	1.523 ± 0.127	2.027 ± 0.142	1.321 ± 0.249			
	<i>Clock</i>	Taar5 +/+	2.71E-02	1.332 ± 0.162	2.065 ± 0.226	0.832 ± 0.318	9.77E-03	3.13E-01	1.92E-02
		Taar5 -/-	5.56E-01	1.883 ± 0.243	2.134 ± 0.274	1.197 ± 0.955			
	<i>Nr1d1</i>	Taar5 +/+	1.23E-04	0.556 ± 0.052	0.93 ± 0.072	1.006 ± 0.198	1.74E-02	7.88E-01	5.11E-04
		Taar5 -/-	3.46E-02	0.755 ± 0.077	0.984 ± 0.08	-1.193 ± 0.364			
	<i>Cry1</i>	Taar5 +/+	9.27E-01	1.372 ± 0.108	1.443 ± 0.15	1.032 ± 2.197	1.61E-01	5.40E-01	1.51E-02
		Taar5 -/-	1.00E-03	1.104 ± 0.162	1.575 ± 0.162	-0.201 ± 0.385			
	<i>Per2</i>	Taar5 +/+	4.14E-02	0.829 ± 0.072	1.16 ± 0.1	1.373 ± 0.31	3.05E-02	6.25E-01	7.66E-03
		Taar5 -/-	2.17E-02	0.498 ± 0.107	0.697 ± 0.106	-0.699 ± 0.61			
Figure 1C: Liver Gene Expression	<i>Amt1</i>	Taar5 +/+	5.81E-03	0.519 ± 0.076	0.936 ± 0.102	0.182 ± 0.273	9.84E-01	6.46E-01	6.64E-02
		Taar5 -/-	2.56E-03	0.614 ± 0.112	1.006 ± 0.113	-0.075 ± 0.304			
	<i>Cry1</i>	Taar5 +/+	1.15E-02	0.471 ± 0.099	0.951 ± 0.132	0.183 ± 0.309	9.67E-01	6.30E-01	1.19E-01
		Taar5 -/-	2.06E-02	0.552 ± 0.143	0.945 ± 0.141	-0.06 ± 0.391			
	<i>Cry2</i>	Taar5 +/+	3.52E-01	0.877 ± 0.248	1.504 ± 0.351	1.018 ± 0.56	7.41E-01	4.80E-01	1.54E-01
		Taar5 -/-	4.72E-01	0.88 ± 0.363	1.186 ± 0.361	-0.053 ± 1.283			
	<i>Per1</i>	Taar5 +/+	3.82E-03	0.669 ± 0.051	0.909 ± 0.069	0.507 ± 0.313	1.23E-01	5.31E-01	7.81E-02
		Taar5 -/-	3.24E-01	0.862 ± 0.074	0.999 ± 0.08	0.781 ± 0.541			
	<i>Per2</i>	Taar5 +/+	2.80E-02	0.602 ± 0.086	0.979 ± 0.116	0.203 ± 0.344	6.76E-01	6.12E-01	1.54E-01
		Taar5 -/-	1.20E-02	0.594 ± 0.125	0.947 ± 0.121	-0.207 ± 0.386			
Figure 1D: Adipose Tissue Gene Expression	<i>Amt1</i>	Taar5 +/+	5.01E-02	0.957 ± 0.052	1.114 ± 0.076	-0.5 ± 0.46	7.46E-01	2.13E-02	2.14E-02
		Taar5 -/-	2.70E-04	1 ± 0.078	1.468 ± 0.074	-0.222 ± 0.197			
	<i>Per1</i>	Taar5 +/+	1.53E-03	2.093 ± 0.241	3.37 ± 0.35	-0.199 ± 0.258	1.28E-01	8.66E-01	6.56E-02
		Taar5 -/-	9.96E-01	2.825 ± 0.358	2.857 ± 0.331	-0.585 ± 13.132			
	<i>Per2</i>	Taar5 +/+	5.72E-05	3.044 ± 0.433	5.964 ± 0.629	-0.217 ± 0.203	4.39E-01	9.09E-01	3.32E-02
		Taar5 -/-	3.74E-01	3.164 ± 0.642	4.201 ± 0.612	-0.218 ± 0.732			
	<i>Prdm16</i>	Taar5 +/+	7.83E-01	1.388 ± 0.321	1.609 ± 0.46	-0.733 ± 2.018	2.42E-02	3.88E-01	6.89E-03
		Taar5 -/-	1.25E-01	2.842 ± 0.476	4.218 ± 0.509	-1.292 ± 0.373			
	<i>Ucp1</i>	Taar5 +/+	6.30E-02	0.822 ± 0.048	1.003 ± 0.066	1.065 ± 0.385	1.15E-01	9.64E-01	3.39E-01
		Taar5 -/-	7.15E-02	0.733 ± 0.072	0.908 ± 0.082	0.564 ± 0.404			
Figure S1A: Circulating Metabolites	<i>Plasma Choline</i>	Taar5 +/+	4.90E-02	36.34 ± 1.937	43.346 ± 2.69	0.955 ± 0.397	3.63E-01	9.33E-01	1.28E-01
		Taar5 -/-	1.25E-01	39.974 ± 2.855	47.106 ± 3.279	1.047 ± 0.383			
	<i>Plasma L-Carnitine</i>	Taar5 +/+	8.53E-01	24.784 ± 0.781	25.398 ± 1.085	0.467 ± 1.832	9.77E-02	7.70E-01	5.40E-01
		Taar5 -/-	8.06E-01	26.771 ± 1.17	27.576 ± 1.242	0.338 ± 1.563			
	<i>Plasma γ-Butyrobetaine</i>	Taar5 +/+	5.02E-01	0.49 ± 0.029	0.53 ± 0.04	0.898 ± 1.027	6.30E-01	7.86E-01	6.70E-07
		Taar5 -/-	9.76E-01	0.528 ± 0.043	0.539 ± 0.045	-1.289 ± 4.027			
	<i>Plasma TMA</i>	Taar5 +/+	5.33E-02	43.982 ± 3.752	54.949 ± 5.213	1.226 ± 0.491	4.46E-01	5.55E-01	1.11E-01
		Taar5 -/-	4.73E-02	48.983 ± 5.53	67.433 ± 5.893	-1.358 ± 0.314			
	<i>Plasma TMAO</i>	Taar5 +/+	8.26E-01	4.139 ± 0.293	4.409 ± 0.41	0.554 ± 1.549	9.24E-01	6.91E-01	3.46E-01
		Taar5 -/-	2.19E-01	4.124 ± 0.446	4.88 ± 0.454	-0.615 ± 0.657			
Figure S1B: Circulating Hormones & Cytokines	<i>Plasma Insulin</i>	Taar5 +/+	6.30E-01	247.888 ± 17.712	272.029 ± 24.564	0.482 ± 1.057	2.86E-01	7.20E-01	2.67E-01
		Taar5 -/-	1.97E-01	276.739 ± 26.221	331.367 ± 29.375	1.213 ± 0.47			
	<i>Plasma PYY</i>	Taar5 +/+	5.41E-01	99.806 ± 6.392	111.17 ± 9.13	-1.56 ± 0.788	4.42E-01	7.12E-01	3.29E-01
		Taar5 -/-	3.43E-01	87.449 ± 9.463	99.271 ± 9.243	-0.614 ± 0.899			
	<i>Plasma Leptin</i>	Taar5 +/+	9.39E-01	3758.578 ± 368.525	3912.379 ± 520.425	-0.224 ± 3.393	3.79E-02	7.31E-01	5.29E-01
		Taar5 -/-	5.79E-01	4704.115 ± 545.559	5435.666 ± 593.184	0.511 ± 0.758			
	<i>Plasma Ghrelin</i>	Taar5 +/+	3.87E-02	4.127 ± 2.019	10.62 ± 2.855	-0.268 ± 0.44	9.26E-01	6.46E-01	2.13E-01
		Taar5 -/-	6.43E-01	3.689 ± 3.025	7.185 ± 3.102	-0.073 ± 0.944			
	<i>Plasma Active GLP-1</i>	Taar5 +/+	3.40E-01	0.119 ± 0.015	0.157 ± 0.022	-0.789 ± 0.56	5.27E-01	8.90E-01	2.74E-05
		Taar5 -/-	2.40E-01	0.1 ± 0.022	0.135 ± 0.025	1.287 ± 0.642			
	<i>Plasma Glucagon</i>	Taar5 +/+	5.53E-02	17.512 ± 2.422	23.735 ± 3.481	-0.774 ± 0.541	8.05E-02	2.59E-01	3.27E-02
		Taar5 -/-	4.32E-02	20.612 ± 3.585	32.869 ± 3.499	-0.468 ± 0.329			
	<i>Plasma C-Peptide</i>	Taar5 +/+	7.10E-01	22.174 ± 11.509	36.075 ± 16.111	0.014 ± 1.183	6.32E-01	9.96E-01	4.47E-01
		Taar5 -/-	4.78E-01	14.894 ± 17.249	36.615 ± 18.463	-1.421 ± 0.831			
	<i>Plasma MCP-1</i>	Taar5 +/+	3.48E-02	17.669 ± 3.13	28.406 ± 4.473	-0.553 ± 0.408	3.39E-01	6.61E-01	1.03E-02
		Taar5 -/-	9.20E-03	24.668 ± 4.633	43.882 ± 4.805	0.159 ± 0.257			
	<i>Plasma IL-6</i>	Taar5 +/+	3.72E-02	14.572 ± 4.767	30.045 ± 6.839	-0.689 ± 0.429	4.48E-01	6.88E-01	8.54E-02
		Taar5 -/-	1.99E-02	22.187 ± 7.057	49.695 ± 7.476	0.309 ± 0.268			
	<i>Plasma TNFα</i>	Taar5 +/+	3.32E-02	4.554 ± 0.25	5.526 ± 0.349	0.965 ± 0.368	2.43E-01	4.84E-01	3.88E-02
		Taar5 -/-	4.59E-03	4.898 ± 0.37	6.47 ± 0.405	0.584 ± 0.237			

Table S1.

Cosinor analyses for all circadian data include in this manuscript.

	Gene	Group	Zero Amplitude Test	MESOR	Amplitude	Acrophase	Mesor Test	Amplitude Test
Figure S3: Gut-Microbial Oscillations	<i>Lachnospiraceae</i> UCG-002	Taar5 +/-	5.97E-01	0.006 ± 0.002	0.008 ± 0.002	1.396 ± 1.392	8.17E-01	6.95E-01
		Taar5/-	8.35E-02	0.008 ± 0.002	0.015 ± 0.002	-0.095 ± 0.37		7.67E-15
	<i>Lachnospiraceae</i> UCG-006	Taar5 +/-	1.37E-02	0.002 ± 0	0.004 ± 0.001	-1.315 ± 0.324	9.84E-01	5.01E-01
		Taar5/-	6.36E-01	0.002 ± 0.001	0.002 ± 0.001	1.46 ± 0.899		5.83E-14
	<i>Lachnospiraceae</i> NK4A136	Taar5 +/-	3.30E-01	0.151 ± 0.013	0.18 ± 0.018	1.349 ± 0.656	4.09E-01	5.61E-01
		Taar5/-	1.13E-02	0.177 ± 0.019	0.237 ± 0.019	-0.991 ± 0.349		2.43E-05
	<i>Lactobacillus</i>	Taar5 +/-	4.33E-02	0.047 ± 0.005	0.065 ± 0.007	1.416 ± 0.414	1.53E-01	8.31E-01
		Taar5/-	5.95E-01	0.042 ± 0.008	0.052 ± 0.009	1.258 ± 0.803		5.14E-01
	<i>Desulfovibrio</i>	Taar5 +/-	3.62E-01	0.017 ± 0.001	0.02 ± 0.002	1.312 ± 0.629	2.25E-01	9.14E-01
		Taar5/-	2.06E-01	0.02 ± 0.002	0.022 ± 0.002	-0.829 ± 0.802		8.89E-13
	<i>Bacteroides</i>	Taar5 +/-	1.27E-01	0.003 ± 0.001	0.005 ± 0.001	0.053 ± 0.658	8.12E-02	3.38E-01
		Taar5/-	1.50E-01	0.005 ± 0.001	0.008 ± 0.001	1.411 ± 0.35		1.80E-15
	<i>Dubosiella</i>	Taar5 +/-	1.75E-02	0.024 ± 0.006	0.04 ± 0.008	1.04 ± 0.538	3.71E-02	3.41E-01
		Taar5/-	6.40E-01	0.043 ± 0.009	0.054 ± 0.009	-0.947 ± 0.88		1.09E-07
	<i>Faecalibaculum</i>	Taar5 +/-	1.04E-01	0.435 ± 0.021	0.494 ± 0.029	1.521 ± 0.517	1.28E-01	4.70E-01
		Taar5/-	2.03E-02	0.374 ± 0.031	0.484 ± 0.032	-1.166 ± 0.303		4.97E-03
	<i>Colidextribacter</i>	Taar5 +/-	2.04E-01	0.009 ± 0.001	0.01 ± 0.001	0.043 ± 0.725	5.77E-01	5.24E-01
		Taar5/-	1.78E-01	0.009 ± 0.001	0.011 ± 0.001	-0.584 ± 0.47		7.46E-16
	<i>Odoribacter</i>	Taar5 +/-	3.48E-01	0.003 ± 0	0.004 ± 0.001	-1.482 ± 0.939	3.29E-01	7.73E-01
		Taar5/-	2.71E-01	0.004 ± 0.001	0.005 ± 0.001	0.626 ± 0.436		1.18E-13
	<i>Butyrivibrio</i>	Taar5 +/-	5.02E-02	0.001 ± 0	0.002 ± 0	1.475 ± 0.403	5.94E-01	6.72E-01
		Taar5/-	8.12E-01	0.001 ± 0	0.002 ± 0	0.616 ± 1.401		4.78E-16
	<i>Alistipes</i>	Taar5 +/-	5.71E-01	0.012 ± 0.001	0.014 ± 0.001	1.387 ± 0.909	1.23E-01	3.38E-01
		Taar5/-	1.21E-02	0.015 ± 0.002	0.02 ± 0.002	1.494 ± 0.309		1.30E-09
	<i>Turicibacter</i>	Taar5 +/-	4.53E-02	0.02 ± 0.003	0.03 ± 0.004	0.005 ± 0.382	1.57E-01	7.19E-01
		Taar5/-	3.36E-01	0.025 ± 0.004	0.032 ± 0.004	0.363 ± 0.609		7.79E-10
	<i>Muribaculum</i>	Taar5 +/-	5.26E-03	0.015 ± 0.001	0.019 ± 0.001	-0.921 ± 0.3	4.95E-02	5.29E-01
		Taar5/-	6.65E-01	0.018 ± 0.001	0.019 ± 0.001	0.463 ± 0.91		2.37E-14
	<i>Coriobacteriaceae</i> UCG-002	Taar5 +/-	1.51E-01	0.001 ± 0	0.001 ± 0	1.38 ± 0.423	2.81E-01	6.11E-01
		Taar5/-	1.23E-01	0 ± 0	0.001 ± 0	-0.737 ± 0.877		1.45E-16
	<i>Acetatifactor</i>	Taar5 +/-	3.19E-01	0 ± 0	0 ± 0	-0.586 ± 0.938	1.00E-02	8.33E-02
		Taar5/-	1.92E-02	0.001 ± 0	0.002 ± 0	0.039 ± 0.259		4.02E-17
	<i>A2</i>	Taar5 +/-	6.79E-01	0.003 ± 0.001	0.004 ± 0.001	-0.549 ± 1.096	5.63E-02	8.03E-01
		Taar5/-	5.48E-01	0.005 ± 0.001	0.006 ± 0.001	1.472 ± 0.794		6.74E-10
Figure 4A: Circulating Metabolites	<i>Helicobacter</i>	Taar5 +/-	3.17E-02	0.108 ± 0.009	0.144 ± 0.012	-1.283 ± 0.34	5.14E-01	9.05E-01
		Taar5/-	1.41E-01	0.117 ± 0.013	0.143 ± 0.013	-1.001 ± 0.539		5.81E-06
	<i>Parabacteroides</i>	Taar5 +/-	5.90E-02	0.005 ± 0.001	0.007 ± 0.001	-1.418 ± 0.551	5.54E-01	4.22E-01
		Taar5/-	2.64E-01	0.005 ± 0.001	0.008 ± 0.001	-0.022 ± 0.491		1.18E-15
	<i>Anaerotruncus</i>	Taar5 +/-	8.75E-01	0.001 ± 0	0.001 ± 0	-0.903 ± 1.74	8.81E-01	4.83E-01
		Taar5/-	2.18E-02	0.001 ± 0	0.002 ± 0	1.261 ± 0.302		2.76E-18
	<i>Plasma Choline</i>	WT-CutC	5.24E-03	48.996 ± 1.613	59.468 ± 2.293	0.405 ± 0.217	1.46E-03	7.74E-01
		ΔCutC	4.47E-07	38.449 ± 2.182	49.649 ± 2.079	0.498 ± 0.186		9.62E-03
	<i>Plasma L-Carnitine</i>	WT-CutC	3.97E-01	37.502 ± 1.209	39.803 ± 1.709	-1.294 ± 0.743	3.91E-01	3.99E-01
		ΔCutC	7.91E-03	39.771 ± 1.636	45.211 ± 1.558	0.511 ± 0.286		7.81E-02
	<i>Plasma Betaine</i>	WT-CutC	1.79E-02	147.098 ± 7.387	177.373 ± 10.633	1.228 ± 0.339	2.36E-01	9.60E-01
		ΔCutC	3.16E-02	135.295 ± 9.995	163.756 ± 9.522	1.138 ± 0.335		3.08E-02
Figure 4B: Olfactory Bulb Gene Expression	<i>Plasma γ-Butyrobetaine</i>	WT-CutC	8.38E-04	1.605 ± 0.073	2.062 ± 0.102	-0.174 ± 0.229	6.75E-01	4.08E-01
		ΔCutC	1.28E-02	1.66 ± 0.099	1.959 ± 0.094	-0.565 ± 0.314		8.52E-03
	<i>Plasma TMA</i>	WT-CutC	1.41E-02	2.315 ± 0.217	3.654 ± 0.312	1.415 ± 0.226	1.48E-03	3.21E-01
		ΔCutC	2.03E-01	0.984 ± 0.294	1.298 ± 0.28	0.583 ± 0.894		7.76E-03
	<i>Plasma TMAO</i>	WT-CutC	8.91E-03	9.425 ± 0.694	14.543 ± 1	1.021 ± 0.188	7.05E-10	9.04E-02
		ΔCutC	3.78E-03	0.378 ± 0.939	0.584 ± 0.894	-1.162 ± 4.339		1.66E-05
	<i>Arnt1</i>	WT-CutC	1.29E-05	0.655 ± 0.12	1.076 ± 0.169	-0.235 ± 0.401	1.28E-01	5.48E-01
		ΔCutC	1.29E-03	0.951 ± 0.16	1.759 ± 0.151	0.129 ± 0.187		1.94E-02
	<i>Clock</i>	WT-CutC	5.92E-02	0.827 ± 0.124	1.214 ± 0.177	0.528 ± 0.447	6.89E-01	6.55E-01
		ΔCutC	6.50E-03	0.766 ± 0.167	1.377 ± 0.159	1.117 ± 0.261		3.07E-02
	<i>Nr1d1</i>	WT-CutC	1.90E-01	0.998 ± 0.232	1.484 ± 0.325	-0.965 ± 0.684	4.61E-01	6.35E-01
		ΔCutC	4.22E-02	1.294 ± 0.315	2.208 ± 0.3	0.842 ± 0.328		1.64E-01
Figure 4C: Circulating Cytokines, Hormones, & Metabolites	<i>Cry1</i>	WT-CutC	1.57E-01	0.49 ± 0.28	0.761 ± 0.391	-0.2 ± 1.484	1.52E-01	5.21E-01
		ΔCutC	3.82E-02	1.105 ± 0.38	2.393 ± 0.362	0.589 ± 0.281		1.70E-01
	<i>Per1</i>	WT-CutC	7.50E-01	1.42 ± 0.203	1.657 ± 0.292	1.034 ± 1.188	4.31E-01	7.79E-01
		ΔCutC	2.86E-01	1.283 ± 0.274	1.684 ± 0.261	-1.169 ± 0.652		2.27E-01
	<i>Per2</i>	WT-CutC	7.06E-01	1.095 ± 0.49	1.588 ± 0.679	-0.514 ± 1.435	5.52E-01	4.24E-01
		ΔCutC	6.56E-02	1.594 ± 0.663	3.301 ± 0.632	0.477 ± 0.37		1.65E-01
	<i>Plasma IL-1β</i>	WT-CutC	5.63E-01	0.162 ± 0.027	0.205 ± 0.038	-0.36 ± 0.88	6.50E-01	8.98E-01
		ΔCutC	1.95E-01	0.145 ± 0.037	0.217 ± 0.039	1.491 ± 0.464		5.59E-04
	<i>Plasma IL-2</i>	WT-CutC	4.21E-03	1.342 ± 0.102	1.936 ± 0.141	1.189 ± 0.25	9.45E-02	8.14E-01
		ΔCutC	5.82E-04	1.562 ± 0.145	2.245 ± 0.152	-1.184 ± 0.202		2.45E-05
	<i>Plasma IL-33</i>	WT-CutC	4.21E-03	1.675 ± 0.128	2.417 ± 0.176	1.189 ± 0.25	9.45E-02	8.14E-01
		ΔCutC	5.82E-04	1.95 ± 0.181	2.802 ± 0.19	-1.184 ± 0.202		3.06E-05
	<i>Plasma Tryptophan</i>	WT-CutC	8.16E-06	256.104 ± 7.029	307.892 ± 9.994	-1.45 ± 0.191	7.59E-01	6.53E-01
		ΔCutC	3.92E-03	252.056 ± 9.511	290.166 ± 9.06	-1.382 ± 0.238		2.25E-02
Figure 4C: Circulating Cytokines, Hormones, & Metabolites	<i>Plasma Leptin</i>	WT-CutC	5.34E-01	25.035 ± 2.737	30.766 ± 3.944	1.006 ± 0.662	5.30E-01	4.47E-01
		ΔCutC	8.20E-04	21.468 ± 3.734	32.134 ± 3.61	0.927 ± 0.335		3.12E-02
	<i>Plasma Active GLP-1</i>	WT-CutC	1.55E-02	0.076 ± 0.01	0.135 ± 0.014	1.272 ± 0.234	9.63E-04	4.56E-01
		ΔCutC	1.00E-01	0.044 ± 0.014	0.063 ± 0.013	0.834 ± 0.685		1.23E-05

Table S1. (continued)

	Gene		Group	Zero Amplitude Test		MESOR		Amplitude		Acrophase	Mesor Test	Amplitude Test
Figure S4A: Brown Adipose Tissue Gene Expression	Arnt1	WT-CutC	1.80E-01	1.653 ± 0.958		2.44 ± 1.37		0.578 ± 1.703		2.39E-01	3.26E-01	1.76E-01
		ΔCutC	1.03E-01	3.207 ± 1.296		6.833 ± 1.235		-0.396 ± 0.341				
	Cry1	WT-CutC	6.33E-02	5.132 ± 1.197		9.024 ± 1.694		-1.326 ± 0.435		9.10E-01	3.75E-01	1.32E-01
		ΔCutC	8.66E-01	5.491 ± 1.62		6.374 ± 1.543		1.193 ± 1.749				
	Per2	WT-CutC	5.80E-02	2.436 ± 0.668		4.608 ± 0.962		1.272 ± 0.428		6.70E-01	7.99E-01	8.42E-02
		ΔCutC	2.02E-01	2.885 ± 0.904		4.606 ± 0.862		-0.126 ± 0.5				
	Pemt	WT-CutC	6.35E-02	2.222 ± 0.436		4.201 ± 0.622		-1.536 ± 0.309		4.09E-01	2.46E-01	1.12E-03
		ΔCutC	7.58E-01	1.925 ± 0.595		2.225 ± 0.581		-0.804 ± 1.881				
	Pdk4	WT-CutC	3.82E-02	0.747 ± 0.227		1.118 ± 0.321		0.2 ± 0.867		2.83E-01	9.45E-01	3.94E-01
		ΔCutC	5.39E-01	1.156 ± 0.31		1.587 ± 0.294		0.684 ± 0.701				
Figure S4B: Circulating Cytokines & Metabolites	Plasma INF-γ	WT-CutC	1.91E-02	0.451 ± 0.048		0.685 ± 0.067		0.84 ± 0.298		3.33E-01	5.40E-01	4.36E-04
		ΔCutC	1.09E-01	0.391 ± 0.068		0.531 ± 0.071		-1.349 ± 0.466				
	Plasma MCP-1	WT-CutC	5.63E-01	0.053 ± 0.009		0.068 ± 0.013		-0.36 ± 0.88		5.62E-01	5.09E-01	5.52E-07
		ΔCutC	1.95E-01	0.048 ± 0.012		0.071 ± 0.013		1.491 ± 0.464				
	Plasma Indole Acetic Acid	WT-CutC	1.48E-03	0.903 ± 0.052		1.145 ± 0.073		0.163 ± 0.306		2.08E-01	9.01E-01	1.86E-03
		ΔCutC	3.94E-02	0.853 ± 0.071		1.062 ± 0.067		-0.548 ± 0.322				
	Plasma Phenylacetic Acid	WT-CutC	1.51E-01	1.224 ± 0.149		1.49 ± 0.213		1.551 ± 0.783		4.15E-06	2.35E-02	4.05E-03
		ΔCutC	1.06E-05	2.654 ± 0.202		4.068 ± 0.192		-0.932 ± 0.136				
	Plasma Phenylacetylglutamine	WT-CutC	6.27E-03	0.913 ± 0.166		1.388 ± 0.238		1.432 ± 0.488		7.37E-06	1.82E-01	6.91E-05
		ΔCutC	6.54E-05	2.573 ± 0.225		3.992 ± 0.214		-1.049 ± 0.151				

Table S1. (continued)

Additional information

Author Contributions

Conceptualization, K.M. and J.M.B.; methodology, K.M., W.J.M., D.O., A.L.B., T.J., A.C.B., A.J.H., S.D., M.M., N.M., V.V., L.J.O., X.Y., D.M.Y., N.Z., R.H., R.B., P.L., D.L., A.M.H., N.S., M.D., J.A.B., and G.R.S., Z.W.; investigation, K.M., W.J.M., D.O., A.L.B., T.J., A.C.B., A.J.H., S.D., M.M., N.M., V.V., L.J.O., X.Y., D.M.Y., N.Z., R.H., R.B., P.L., D.L., N.S., M.D., J.A.B., and Z.W.; validation, K.M., W.J.M., A.C.B., A.J.H., S.D., M.M., L.J.O., R.B., P.L., D.L., N.S., M.D., J.A.B., and Z.W.; formal analysis, K.M., W.J.M., N.S., T.G., G.R.S., Z.W., J.M.B.; writing – original draft, K.M. and J.M.B.; writing – reviewing and editing, K.M., W.J.M., D.O., A.L.B., T.J., A.C.B., A.J.H., S.D., M.M., N.M., V.V., L.J.O., X.Y., D.M.Y., N.Z., R.H., R.B., P.L., D.L., A.M.H., N.S., M.D., J.A.B., G.R.S., and Z.W.; funding acquisition, J.M.B.; supervision, J.M.B.

Abbreviations used

Arntl, basic helix-loop-helix ARNT like 1 or Bmal1; BAT, brown adipose tissue; CKD, chronic kidney disease; *Cry1*, cryptochrome 1; *Cry2*, cryptochrome 2; CVD, cardiovascular disease; EIF2AK3, eukaryotic translation initiation factor 2-alpha kinase 3; ER, endoplasmic reticulum; FMO3, flavin containing monooxygenase 3; GPCR, G protein-coupled receptor; IFN γ , interferon gamma; IL, interleukin; MCP-1, monocyte chemoattractant protein-1; NF κ B, nuclear factor κ B; NLRP3, nucleotide-binding domain, leucine-rich-containing family, pyrin domain-containing-3; *Nr1d1*, nuclear receptor subfamily 1 group D member 1 or REV-ERB α ; Pemt, phosphatidylethanolamine *N*-methyltransferase; PERK, eukaryotic translation initiation factor 2-alpha kinase 3; *Per1*, period 1; *Per2*, period 2; *Taar5*, trace amine associated receptor 5; TMA, trimethylamine; TMAO, trimethylamine oxide; WAT, white adipose tissue; ZT, zeitgeber time.

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Reviewer #1 (Public review):

Summary:

This study focuses on the bacterial metabolite TMA, generated from dietary choline. These authors and others have previously generated foundational knowledge about the TMA metabolite TMAO, and its role in metabolic disease. This study extends those findings to test whether TMAO's precursor, TMA, and its receptor TAAR5 are also involved and necessary for some of these metabolic phenotypes. They find that mice lacking the host TMA receptor (Taar5^{-/-}) have altered circadian rhythms in gene expression, metabolic hormones, gut microbiome composition, and olfactory and innate behavior. In parallel, mice lacking bacterial TMA production or host TMA oxidation have altered circadian rhythms.

Strengths:

These authors use state-of-the-art bacterial and murine genetics to dissect the roles of TMA, TMAO, and their receptor in various metabolic outcomes (primarily measuring plasma and tissue cytokine/gene expression). They also follow a unique and unexpected behavioral/olfactory phenotype. Statistics are impeccable.

Weaknesses:

Enthusiasm for the manuscript is dampened by some ambiguous writing and the presentation of ideas in the introduction, both of which could easily be improved upon revision.

<https://doi.org/10.7554/eLife.107037.1.sa2>

Reviewer #2 (Public review):

Summary:

In the manuscript by Mahen et al., entitled "Gut Microbe-Derived Trimethylamine Shapes Circadian Rhythms Through the Host Receptor TAAR5," the authors investigate the interplay between a host G protein-coupled receptor (TAAR5), the gut microbiota-derived metabolite trimethylamine (TMA), and the host circadian system. Using a combination of genetically engineered mouse and bacterial models, the study demonstrates a link between microbial signaling and circadian regulation, particularly through effects observed in the olfactory system. Overall, this manuscript presents a novel and valuable contribution to our understanding of host-microbe interactions and circadian biology. However, several sections would benefit from improved clarity, organization, and mechanistic depth to fully support the authors' conclusions.

Strengths:

- (1) The manuscript addresses an important and timely topic in host-microbe communication and circadian biology.
- (2) The studies employ multiple complementary models, e.g., Taar5 knockout mice, microbial mutants, which enhance the depth of the investigation.
- (3) The integration of behavioral, hormonal, microbial, and transcript-level data provides a multifaceted view of the observed phenotype.
- (4) The identification of olfactory-linked circadian changes in the context of gut microbes adds a novel perspective to the field.

Weaknesses:

While the manuscript presents compelling data, several weaknesses limit the clarity and strength of the conclusions.

(1) The presentation of hormonal, cytokine, behavioral, and microbiome data would benefit from clearer organization, more detailed descriptions, and functional grouping to aid interpretation.

(2) Some transitions-particularly from behavioral to microbiome data-are abrupt and would benefit from better contextual framing.

(3) The microbial rhythmicity analyses lack detail on methods and visualization, and the sequencing metadata (e.g., sample type, sex, method) are not clearly stated.

(4) Several figures are difficult to interpret due to dense layouts or vague legends, and key metabolites and gene expression comparisons are either underexplained or not consistently assessed across models.

(5) Finally, while the authors suggest a causal role for TAAR5 and its ligand in circadian regulation, the current data remain correlative; mechanistic experiments or stronger disclaimers are needed to support these claims.

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Reviewer #3 (Public review):

Summary:

Deletion of the TMA-sensor TAAR5 results in circadian alterations in gene expression, particularly in the olfactory bulb, plasma hormones, and neurobehaviors.

Strengths:

Genetic background was rigorously controlled.

Comprehensive characterization.

Weaknesses:

The weaknesses identified by this reviewer are minor.

Overall, the studies are very nicely done. However, despite careful experimentation, I note that even the controls vary considerably in their gene expression, etc, across time (eg, compare control graphs for Cry 1 in IB, 4B). It makes me wonder how inherently noisy these measurements are. While I think that the overall point that the Taar5 KO shows circadian changes is robust, future studies to dissect which changes are reproducible over the noise would be helpful.

Impact:

These data add to the growing literature pointing to a role for the TMA/TMAO pathway in olfaction and neurobehavioral.

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